

Thus, the effects of multiple hormones (ADH, angiotensin II, aldosterone) on sodium and fluid retention and water intake result in an increase in blood volume, which helps to maintain blood pressure. These mechanisms are important in the physiological responses to hemorrhage or dehydration; various aspects are also implicated in elevation of blood pressure in some forms of hypertension.



The **diving reflex** is a specialized mechanism for regulation of blood pressure and heart rate. This reflex is an adaptation for conservation of oxygen in diving mammals, allowing protracted stays underwater without breathing. During diving, the heart rate is slowed by increased vagal activity, while pressure is maintained by arterial vasoconstriction. Thus, blood flow to vital organs is maintained, while blood flow to much of the body besides the coronary and cerebral circulations is reduced; the work of the heart is also reduced. Although the diving reflex is weaker in humans, its effects can still be observed when the face is submerged in cold water and breath is held. Receptors in the face and nasal cavities are stimulated, and reflexive bradycardia and peripheral vasoconstriction occur. The diving reflex is thought to be responsible for the survival of some children after long periods of accidental submersion in cold water.

SPECIAL CIRCULATIONS

Blood flow to various tissues is regulated by local and extrinsic mechanisms, and the importance of these varies among tissues (Fig. 12.8). In addition, there are unique aspects of some circulations. A number of these “special circulations” merit individual consideration.

Cerebral Circulation

The **cerebral circulation** is supplied by the arterial circle of Willis, which is derived from the internal carotid and the vertebral arteries. Because the brain is a vital tissue, housed in a rigid case (the cranium), it is necessary to closely regulate its blood flow. Flow below 35 milliliters per minute (mL/min) causes neuronal dysfunction; high flow may result in cerebral edema and high intracranial pressure, which also produce dysfunction. In a normal individual, cerebral blood flow is maintained at a constant level of 50 mL/min for mean arterial pressures between 50 mm Hg and 150 mm Hg. This level is maintained by autoregulation (myogenic regulation).

Cerebral blood flow is also under the control of arterial PCO_2 , although this blood gas does not vary significantly in a healthy person except during extreme exercise or hyperventilation or hypoventilation. Decreased arterial PCO_2 will produce vasoconstriction and reduced cerebral blood flow; high arterial PCO_2 levels will cause vasodilation and increased flow. Note that this response is to *arterial* PCO_2 . Global cerebral blood flow is not significantly regulated by local metabolic factors (CO_2 , O_2 , K^+ , H^+ , etc.), but metabolic regulation participates



The circle of Willis provides a high degree of collateralization between the large arteries that supply oxygenated blood to the brain. The redundant pathways for blood flow help to ensure adequate blood flow to the brain in case of injury or disease, for example due to stenosis of vessels or blockage due to thrombosis. There is considerable variation in the exact anatomic structure of the circle of Willis between individuals, such that fewer than half of the population has the most common structure illustrated in many textbooks.

in distribution of cerebral blood flow to various parts of the brain, depending on neuronal activity in those regions.

In addition, there are two reflexes that may affect cerebral blood flow:

- **CNS ischemic reflex:** If the vasomotor center in the medulla becomes ischemic, strong sympathetic outflow acts on the heart and peripheral circulation to increase arterial blood pressure above the autoregulatory range for cerebral blood flow, thus increasing blood flow to the brain in a “last-ditch” effort to reverse the ischemia. The cerebral circulation is itself not innervated by sympathetic nerves.
- **Cushing reflex:** An increase in intracranial pressure, usually associated with traumatic head injury, may impede cerebral blood flow. A strong sympathetic outflow will result, raising arterial pressure in an attempt to overcome the high intracranial pressure that is impeding flow. This is also a last-ditch mechanism to preserve cerebral perfusion. High arterial pressure is accompanied by bradycardia, due to activation of arterial baroreceptors.

Coronary Circulation

Like any metabolically active tissue, the myocardium receives its own arterial supply. The **coronary circulation** is fed by the right and left coronary arteries that originate at the base of the aorta (Fig. 12.9). These are also called epicardial arteries because they lie on the surface of the heart (epicardium) and supply branches into the myocardium. These branches form a very extensive microcirculation, supplying oxygen and nutrients and removing metabolites from the highly metabolically active myocardium. The heart has a very large arteriovenous O_2 gradient, even at rest.

The left coronary artery gives rise to the left anterior descending (LAD) artery and the circumflex artery; likewise, the right coronary artery gives rise to several branches. The venous drainage of the heart returns blood to the coronary sinus, which empties into the right atrium. The layers of the heart, from inside to outside, are as follows:

- **Endocardium:** The innermost layer is connective tissue, lined by endothelial cells.

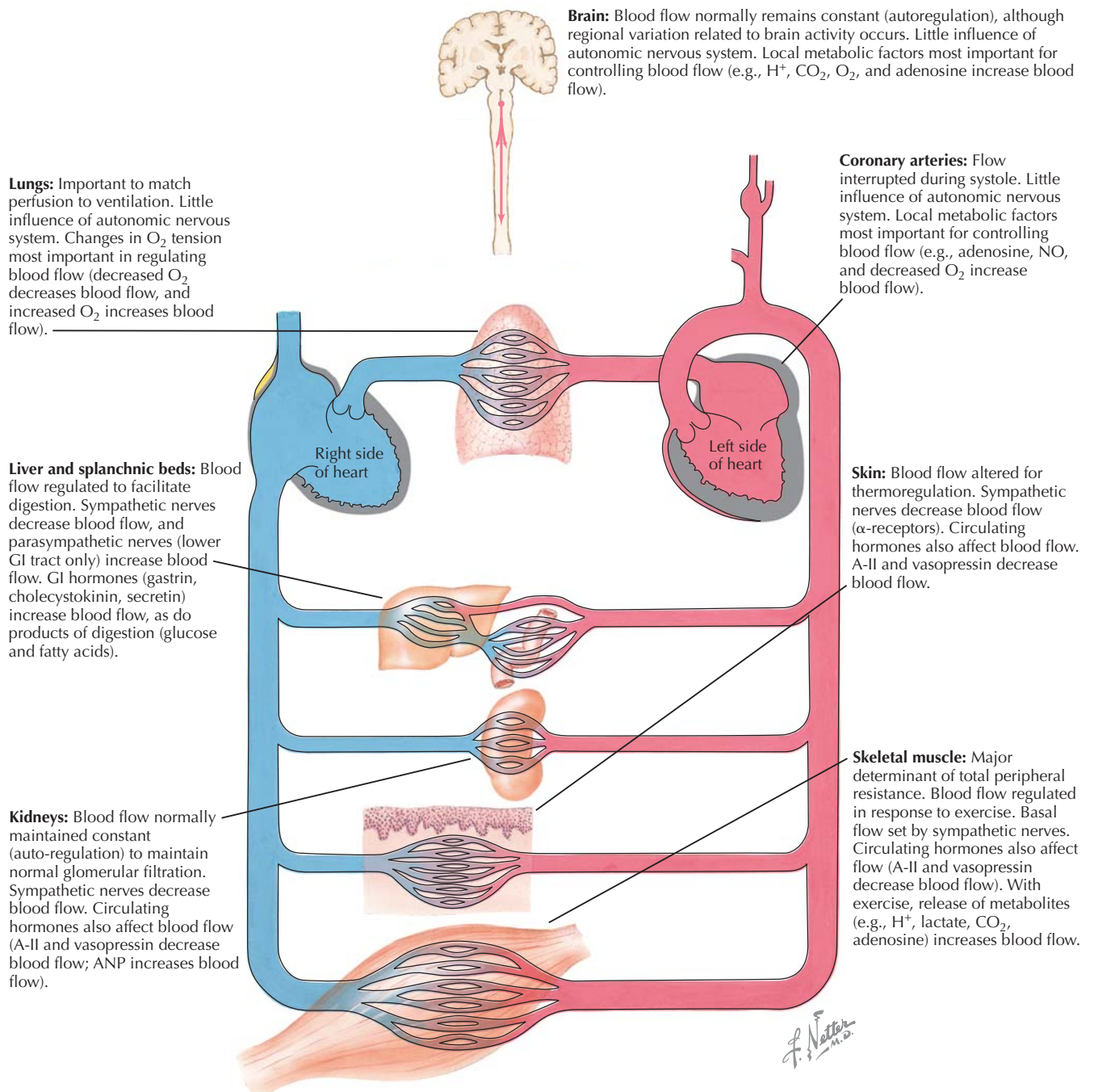


Figure 12.8 Circulation to Special Regions Mechanisms regulating blood flow vary among regions in the systemic circulation, reflecting the physiological functions and needs of the tissues.

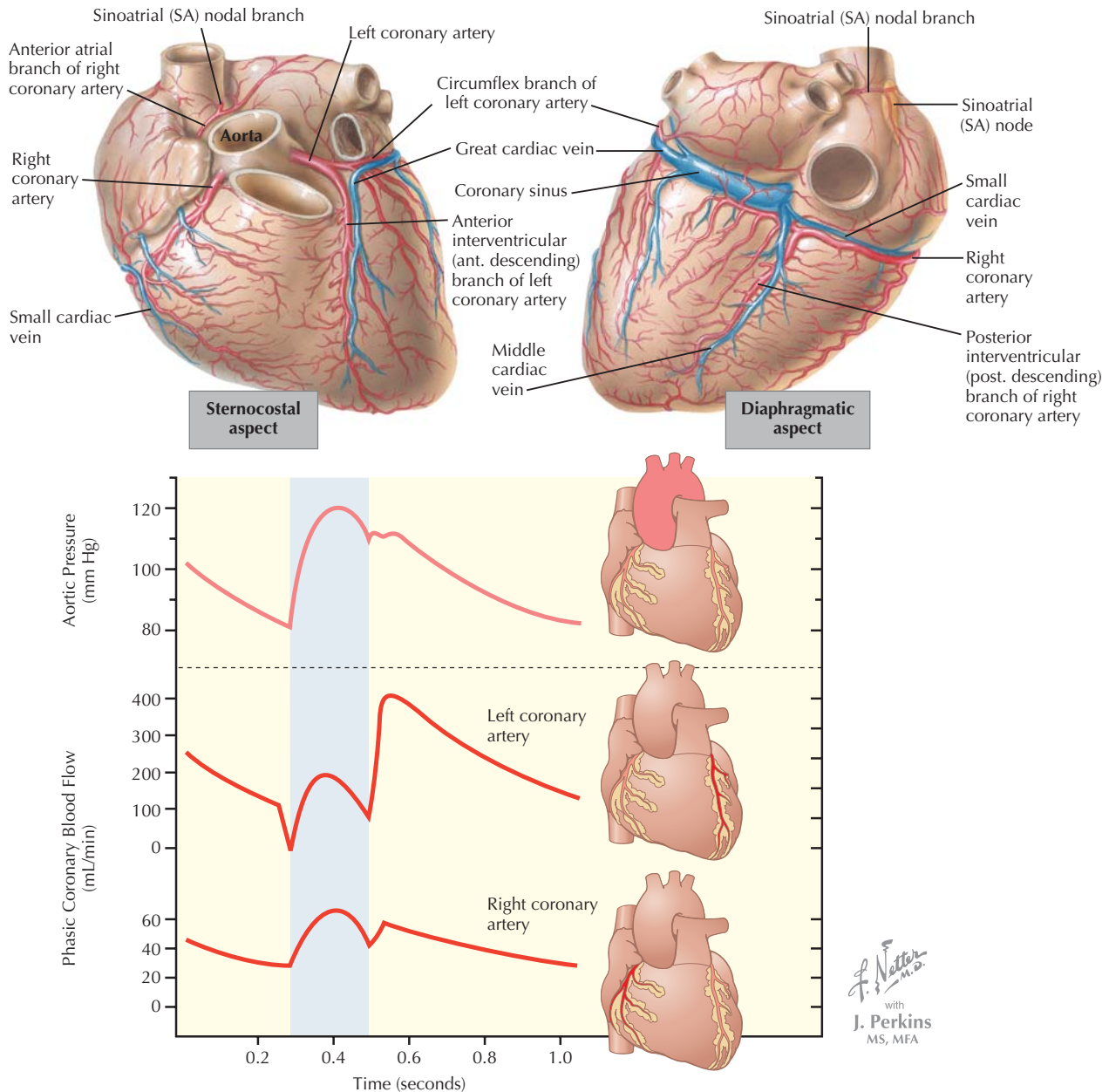


Figure 12.9 Coronary Circulation The coronary arteries supply arterial blood to the muscular wall of the heart. The pressure gradient for flow through the arteries is affected by tissue pressure in the wall of the heart during systole, particularly in the left coronary circulation. Thus, while flow through both right and left coronary arteries is related to aortic pressure, these compressive forces reduce left coronary flow during systole. During diastole, flow is greatly enhanced in the left coronary circulation by the actions of vasodilator metabolites that build up during systole. The metabolite adenosine is believed to be particularly important in this enhancement of coronary flow.

- **Myocardium:** The muscular layer that forms the bulk of the heart wall. The subendocardium is the inner $\frac{1}{3}$ of the myocardium, and the subepicardium is the outer $\frac{1}{3}$ of the myocardium.
- **Epicardium:** The outer, connective tissue layer of the heart. It forms the visceral pericardium. The parietal pericardium is a sac enclosing the heart; the pericardial cavity is the space between the visceral and parietal pericardium.

Flow to the myocardium is affected by factors that are in some ways different from those regulating flow in other circulations. In particular, coronary blood flow is affected greatly by the following:

- Compression of the coronary circulation caused by contraction of myocardium.
- Powerful metabolic vasodilation during diastole.

The pumping action of the left ventricle generates the normal resting arterial pressure of 120/80 mm Hg to supply blood to the systemic circulation. In doing so, the myocardium of the left ventricle, and in particular, the subendocardium, must generate an extravascular tissue pressure that is higher than left ventricular and arterial pressures. This intramyocardial pressure impedes left coronary flow during systole, when the net coronary perfusion pressure is arterial pressure minus the intramyocardial pressure. Thus, left coronary artery flow is low during systole (see Fig. 12.9). During isovolumetric contraction, there is a rapid decrease in flow because tissue pressure has exceeded arterial pressure. During the remainder of systole, the shape of the left coronary flow curve is similar to the aortic pressure curve, but remains low due to high intramyocardial pressure.

With the onset of isovolumetric relaxation, left coronary artery flow increases markedly due to two factors. During systole, when flow is low, metabolites (H^+ , CO_2 , K^+ , prostaglandins, lactic acid, adenosine, and others) accumulate, and O_2 tension falls. These changes produce coronary vasodilation; adenosine is believed to be the major factor involved. The vasodilation, combined with the fall in intramyocardial pressure, results in the large increase in left coronary flow during diastole.

Flow through the right coronary artery is less affected by intramyocardial pressure, because tissue pressure in the right ventricle does not nearly reach that of the left ventricle, and does not exceed arterial pressure. Thus, the shape of the right coronary artery blood flow curve (see Fig. 12.9) resembles the aortic pressure curve, with the highest flow occurring during systole.

Sympathetic nerves innervate coronary arteries and can affect blood flow, but to a lesser extent than in most other circulations. Activation of the sympathetic nervous system increases

the work of the heart, and thus metabolic vasodilation largely overrides sympathetic coronary vasoconstriction. However, it is believed that sympathetic vasoconstriction may limit the extent of metabolic vasodilation, and thus, in a compromised heart (for example, in coronary heart disease), sympathetic stimulation may contribute to ischemia.

Autoregulation of coronary blood flow can be demonstrated experimentally but is a less important factor in normal regulation of coronary flow than metabolic factors or the effects of intramyocardial pressure.

Blood Flow in Exercise and Effects on Specific Circulations

The importance of regional differences in regulatory mechanisms governing blood flow can be gleaned by considering the changes in flow that occur during exercise. Regulation of arterial blood pressure and flow during **dynamic exercise** is a complex process, involving large changes in cardiac output and regional resistances (Fig. 12.10). Dynamic exercise, also called aerobic exercise, involves rhythmic contraction/relaxation of large skeletal muscle groups, for example during jogging, swimming, or aerobics. Young, fit individuals are able to exercise at levels associated with cardiac outputs of 20 to 30 L/min, by increasing both heart rate and stroke volume. Sympathetic nervous system activation results in vasoconstriction and reduced flow (as a proportion of cardiac output) in many, but not all, regional circulations (see Fig. 12.10):

- Coronary blood flow increases, mainly as a result of metabolic vasodilation.
- Skin blood flow is reduced at first, but will eventually rise as a result of thermoregulatory mechanisms.
- Skeletal muscle flow increases dramatically, as constrictor effects of catecholamines are overridden by metabolic vasodilation (and to some extent, activation of β -adrenergic receptors).

Total peripheral resistance is reduced as a result of these changes. Thus, aerobic exercise is associated with high cardiac output and low resistance. Constriction of veins results in increased central venous pressure, augmenting cardiac filling pressure and stroke volume. Arterial blood pressure is elevated during dynamic exercise, but it is mainly systolic pressure (and thus pulse pressure) that rises, due to the large stroke volume. Diastolic pressure may be slightly elevated or slightly reduced, because low peripheral resistance results in a rapid fall in arterial pressure during diastole.

Fetal Circulation

Although the **fetal circulation** is not a special circulation in the same sense as the cerebral and coronary circulations, there

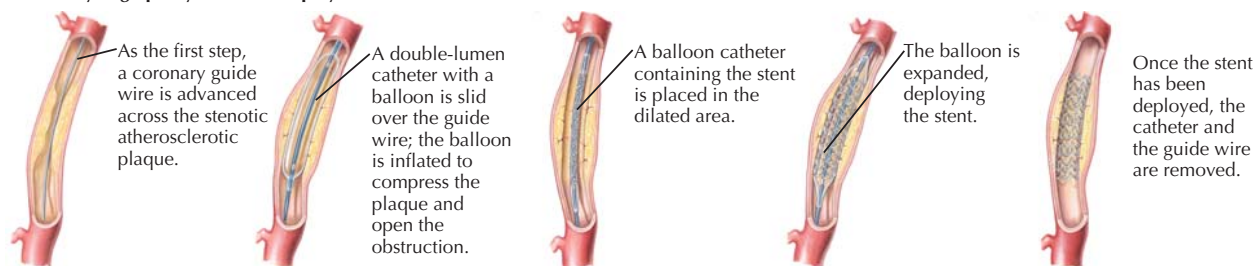
CLINICAL CORRELATE

Myocardial Ischemia

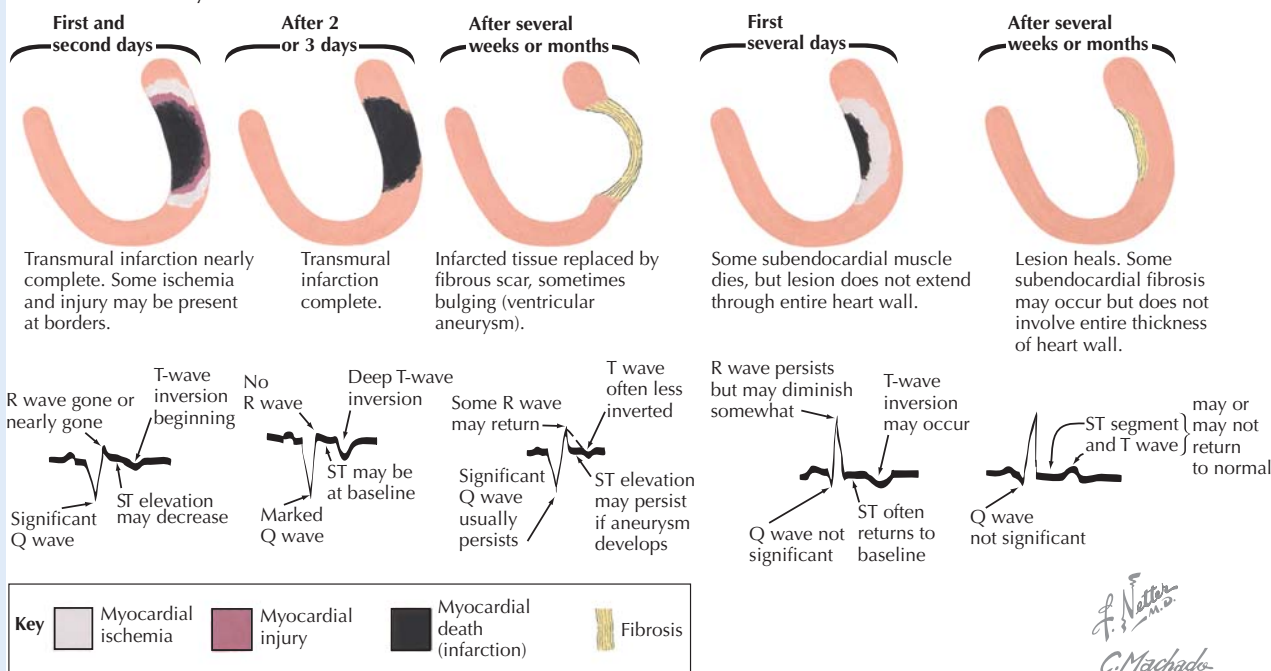
Myocardial ischemia is reduced blood flow to heart muscle, resulting in poor oxygenation. Angina, arrhythmias, myocardial infarction, and sudden death may result. In patients with coronary heart disease, buildup of atherosclerotic plaque in epicardial coronary arteries results in reduced perfusion of the affected vessels. Functionally, ischemia and infarction may result in reduced contractility of the heart, arrhythmias, electrocardiogram (EKG) changes,

wall motion abnormalities, and even ventricular aneurysm. Myocardial infarction is one of the major causes of congestive heart failure. Severe cases of coronary occlusion are treated by angioplasty or coronary artery bypass graft surgery. In bypass surgery, a section of vessel, typically a piece of saphenous vein, is grafted to bypass the blockage. In angioplasty, a balloon catheter is employed to open the occluded artery. To reduce the occurrence of restenosis (reocclusion), a stent is often deployed.

A. Coronary angioplasty and stent deployment



B. Manifestations of myocardial infarction



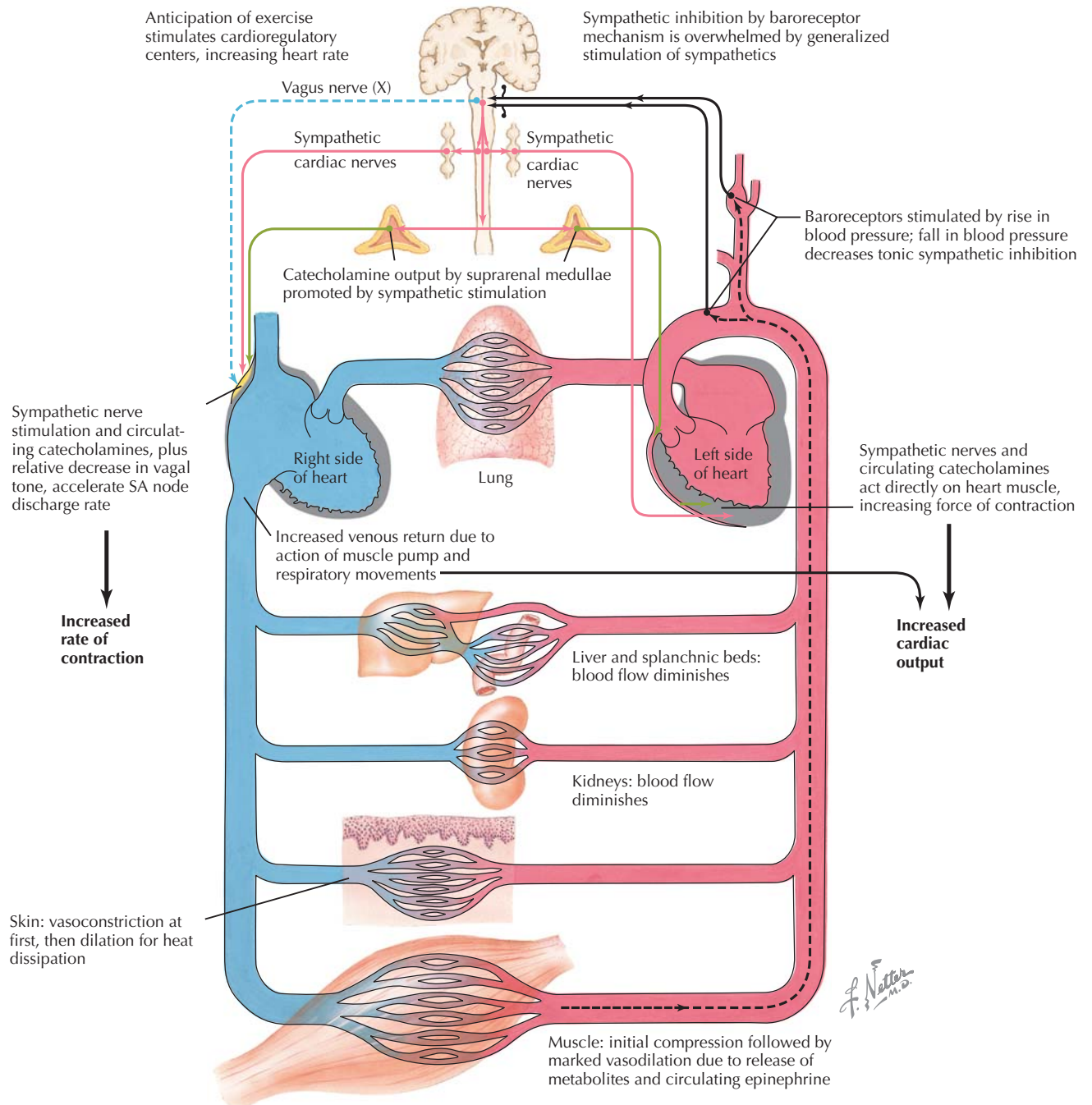


Figure 12.10 Circulatory Response to Exercise Dynamic (aerobic) exercise elicits an integrated circulatory response. The autonomic nervous system is important in the neural regulation of this response, affecting heart rate, contractility, and vascular tone. During exercise, skeletal muscle blood flow is greatly elevated due mainly to the production of vasodilatory metabolites.

Prenatal circulation

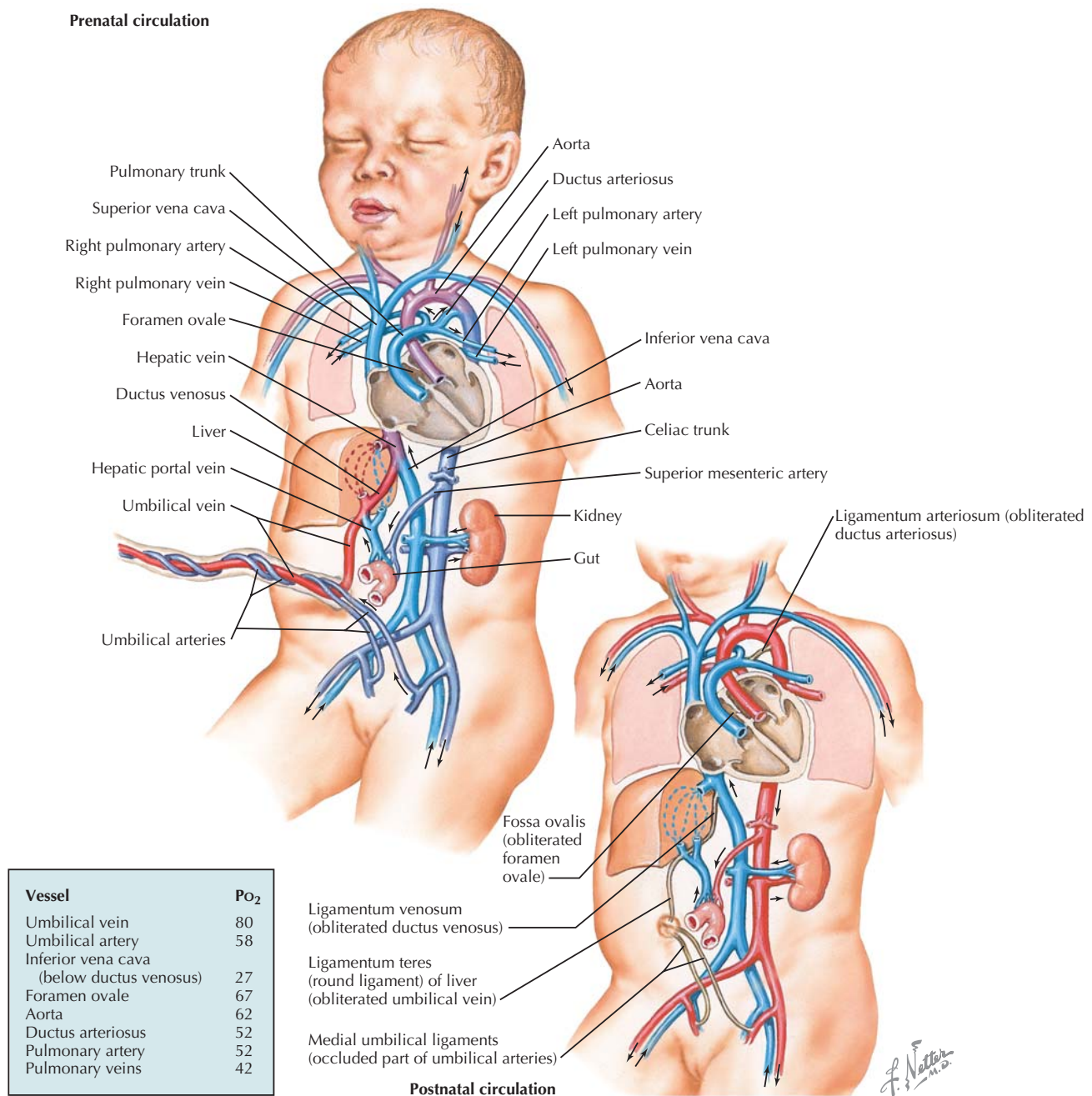


Figure 12.11 Fetal and Neonatal Circulation The fetal circulation is specifically adapted to efficiently exchange gases, nutrients, and wastes through placental circulation. Upon birth, the shunts (foramen ovale, ductus arteriosus, and ductus venosus) close and the placental circulation is disrupted, producing the series circulation of blood through the lungs, left atrium, left ventricle, systemic circulation, right heart, and back to the lungs.

are several significant differences between the prenatal and postnatal circulations that merit discussion. The fetus, enveloped in the amniotic sac and floating in the amniotic fluid, is dependent on the placental circulation for exchange of gases and nutrients. To adapt the circulation to prenatal life, there

are six structures in the fetus not normally seen in adults (Fig. 12.11):

- Two large umbilical arteries, which branch from the systemic arterial circulation and supply the placental

circulation, where gas exchange and nutrient and waste exchange occur.

- The umbilical vein, which returns placental blood to the systemic venous circulation, supplying O₂ and nutrients to the fetus.
- The ductus venosus, a shunt between the umbilical vein and the inferior vena cava. Although most of the placental blood passes through the liver, a fraction of the placental circulation passes directly into the vena cava through the ductus venosus.
- Foramen ovale, a “right-to-left” shunt through which most of the blood from the inferior vena cava flows into the left atrium, bypassing the right heart and the pulmonary circulation.
- Ductus arteriosus, another right-to-left shunt between the pulmonary artery and the aorta. Pulmonary vascular resistance is high in the fetus, due to the collapsed state of the lungs and low oxygen tension in the lungs; 90% of pulmonary arterial blood flows through the ductus arteriosus to the aorta.

These structures all normally close shortly after birth, although anatomical closure, as opposed to functional closure, requires



Unlike the adult, the fetal circulation can be considered a parallel circulation. Much of the right ventricular output is shunted past the lungs to the aorta through the ductus arteriosus, mixing with blood ejected by the left ventricle. Flow through the pulmonary circulation is not critical, since oxygenation of blood does not take place in the lungs in utero. With closure of the foramen ovale and ductus arteriosus shortly after birth, the fetal circulation is transformed to a series circulation.

more time. Inflation of the lungs and inhalation of oxygen-rich air reduces pulmonary vascular resistance, reversing flow through the ductus arteriosus. High oxygen tension initiates closure of the ductus. With increased venous return from the lungs to the left atrium and reduced flow to the right atrium due to occlusion of the placental circulation, left atrial pressure rises above right atrial pressure, closing a valve over the foramen ovale. With closure of the ductus arteriosus and foramen ovale, the “adult” pattern of circulation of blood through the right heart, lungs, left heart, and systemic circulation is established.

Review Questions

CHAPTER 8: OVERVIEW OF THE HEART AND CIRCULATION

- The greatest volume of blood is found in the:
 - systemic arteries.
 - systemic veins.
 - pulmonary arteries.
 - pulmonary veins.
 - chambers of the heart.
- Contraction of which chamber of the heart is best described as a constriction of the chamber resulting in shortening from base to apex?
 - Left atrium
 - Right atrium
 - Left ventricle
 - Right ventricle
 - All of the above
- Of the following parts of the circulation, which receives the smallest blood flow (as a percentage of cardiac output) at rest?
 - Coronary circulation
 - Liver and gastrointestinal tract
 - Skeletal muscle
 - Kidneys
 - Cerebral circulation

CHAPTER 9: CARDIAC ELECTROPHYSIOLOGY

- Rapid phase 3 repolarization of cardiac myocytes during the action potential is mainly associated with:
 - opening of Na^+ channels.
 - reduced conductance of the inwardly rectified K^+ current (i_{K1}).
 - opening of voltage-sensitive slow L-type Ca^{2+} channels.
 - inactivation of Na^+ channels.
 - activation of K^+ channels.
- The electrocardiogram may reveal:
 - abnormalities in cardiac rhythm and conduction.
 - location and extent of myocardial ischemia.
 - orientation of the heart and size of chambers.
 - effects of abnormal electrolyte levels.
 - All of the above.

- Conduction velocity is slowest in the:

- atrial myocytes.
- AV node.
- bundle of His.
- Purkinje fibers.
- ventricular myocytes.

CHAPTER 10: FLOW, PRESSURE, AND RESISTANCE

- Given a resting heart rate of 75 and arterial blood pressure of 130/80 mm Hg, what is the approximate mean arterial pressure (MAP)?
 - 90 mm Hg
 - 97 mm Hg
 - 105 mm Hg
 - 110 mm Hg
 - 115 mm Hg
- The highest pulse pressure in the circulation is observed in the:
 - left ventricle.
 - right ventricle.
 - aorta.
 - pulmonary artery.
 - systemic arterioles.
- Which of the following changes would result in the greatest increase in flow through a tube, assuming that other factors are held constant?
 - Doubling of the pressure gradient between ends of the tube
 - Reducing the viscosity of the fluid by half
 - Doubling the viscosity of the fluid
 - Reducing the length of the tube by half
 - Doubling the radius of the tube
- Which of the following conditions favors laminar flow in a tube?
 - Wide tube diameter
 - Rapid velocity of fluid flow
 - High viscosity of fluid
 - High density of fluid
 - Pulsatility of flow

11. Wall tension in a blood vessel would be increased by:
- A. reduction of vessel radius.
 - B. reduction of hydrostatic pressure within the vessel lumen.
 - C. reduction of hydrostatic pressure in the interstitial space.
 - D. reduction of transmural pressure.
 - E. None of the above.

CHAPTER 11: THE CARDIAC PUMP

12. The beginning of the isovolumetric contraction period for the right ventricle is marked by:

- A. closure of the aortic valve.
- B. closure of the pulmonic valve.
- C. closure of the mitral valve.
- D. closure of the tricuspid valve.
- E. opening of the mitral valve.

13. The correct order of valve opening and closure during a cardiac cycle is:

- A. mitral closure, tricuspid closure, pulmonic opening, aortic opening, aortic closure, pulmonic closure, tricuspid opening, mitral opening.
- B. tricuspid closure, mitral closure, pulmonic opening, aortic opening, aortic closure, pulmonic closure, tricuspid opening, mitral opening.
- C. mitral closure, tricuspid closure, aortic opening, pulmonic opening, aortic closure, pulmonic closure, tricuspid opening, mitral opening.
- D. mitral closure, tricuspid closure, pulmonic opening, aortic opening, pulmonic closure, aortic closure, tricuspid opening, mitral opening.
- E. mitral closure, tricuspid closure, pulmonic opening, aortic opening, aortic closure, pulmonic closure, mitral opening, tricuspid opening.

14. In respiratory sinus arrhythmia, the increase in heart rate during inspiration is initiated by:

- A. reduced stretch of arterial baroreceptors.
- B. increased stretch of arterial baroreceptors.
- C. release of atrial natriuretic peptide.
- D. reduced stretch of atrial stretch receptors.
- E. increased stretch of atrial stretch receptors.

15. Activation of the sympathetic nervous system will produce:

- A. increased heart rate.
- B. arterial vasoconstriction.
- C. constriction of veins.
- D. increased myocardial contractility.
- E. All of the above.

16. An increase in myocardial contractility will result in which of the following changes in the force–velocity relationship?

- A. An increase in the maximum force of contraction without change in V_m
- B. An increase in the maximum force of contraction and an increase in V_m
- C. A decrease in the maximum force of contraction without change in V_m
- D. A decrease in the maximum force of contraction and an increase in V_m
- E. A decrease in the maximum force of contraction and a decrease in V_m

17. When the vascular function curve and cardiac function curves are plotted on the same graph, the point of intersection represents the:

- A. point at which baroreceptor activation occurs.
- B. resting cardiac output and central venous pressure.
- C. normal mean circulatory pressure.
- D. normal mean arterial pressure.
- E. point of maximum contractility.

CHAPTER 12: THE PERIPHERAL CIRCULATION

18. Release of nitric oxide by the endothelial cells in a small artery would be expected to produce all of the following effects **except**:

- A. elevation of cGMP in smooth muscle underlying the endothelium.
- B. inhibition of platelet adhesion to the vascular wall.
- C. elevation of free intracellular Ca^{2+} in the underlying smooth muscle.
- D. vasodilation of the vessel.
- E. increased hydrostatic pressure in downstream capillaries.

19. In an experimental model, the perfusion pressure for an artery is increased. A rise in blood flow through the artery is observed, but despite continued high perfusion pressure, flow falls over a 2-minute period toward the baseline level. This reduction in flow is most likely the result of:

- A. autoregulation of blood flow.
- B. reactive hyperemia.
- C. active hyperemia.
- D. shear stress–induced nitric oxide release.
- E. increased sympathetic nerve activity.

20. Vasodilation is evoked by binding of adrenergic transmitter by which type of vascular adrenergic receptor?

- A. α_1
- B. α_2
- C. β_1
- D. β_2
- E. All of the above

21. Afferent nerve activity from the arterial baroreceptors will be greatest when:
- A. MAP and pulse pressure are both high.
 - B. MAP is high and pulse pressure is low.
 - C. MAP and pulse pressure are both low.
 - D. MAP is low and pulse pressure is high.
 - E. MAP and pulse pressure are in the normal resting range.
22. Left coronary artery blood flow is highest during:
- A. early systole.
 - B. late systole.
 - C. early diastole.
 - D. mid diastole.
 - E. late diastole.

Section **4**

RESPIRATORY PHYSIOLOGY

The components of the respiratory system are the lungs, airways, and the muscles of breathing. The primary function of this system is to facilitate the exchange of oxygen and carbon dioxide between the body and the environment. The respiratory system participates in additional, important physiological processes, including acid–base regulation, temperature regulation, immune function, and metabolic functions, among others. In the following chapters, the basic physiology of ventilation, pulmonary perfusion, and gas transport are covered, as well as the critical role of the respiratory system in acid–base balance.

Chapter 13 **Pulmonary Ventilation and Perfusion and Diffusion of Gases**

Chapter 14 **The Mechanics of Breathing**

Chapter 15 **Oxygen and Carbon Dioxide Transport and Control of Respiration**

Review Questions

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CHAPTER 13

Pulmonary Ventilation and Perfusion and Diffusion of Gases

Respiration is a complex process that begins with ventilation of the lungs and diffusion of gases between the lungs and blood. Meanwhile, the lungs are perfused with blood, which transports gases between lungs and tissues, where the biochemical processes involved in cellular respiration occur.

BLOOD FLOW IN THE LUNGS

The general principles of blood flow are discussed in Section 3, but the systemic and pulmonary circulations are different in some important aspects. Although right ventricular output is the same as left ventricular output (normally about 5 liters per minute [L/min] at rest), pressures in the pulmonary circulation are lower than pressures at equivalent points in the systemic circulation (Fig. 13.1). To accommodate high flow rate at lower pressure, resistance is also low in the pulmonary circulation. This low resistance is a result of the dense microcirculation and shorter vessels in the lungs compared with the systemic circulation.

Passive Control of Pulmonary Resistance

Pulmonary resistance is subject to **passive control** by pulmonary artery pressure and lung volume. Ordinarily, some pulmonary capillaries are collapsed. When **pulmonary artery**

pressure increases, recruitment and distension of pulmonary capillaries occurs. In other words, collapsed capillaries are recruited to the circulation by the higher perfusion pressure, and individual capillaries distend, further reducing resistance. Such recruitment and distension are important in reducing pulmonary vascular resistance during high cardiac output, for example, in exercise. **Lung volume** has different effects on extra-alveolar vessels and alveolar capillaries. As lung volume increases, the associated traction causes distension of extra-alveolar vessels, reducing resistance within them. In contrast, with increased lung volume, alveolar vessels are compressed by the inflation of alveoli. Thus, as lungs are inflated from a very low volume, resistance first falls due to the effects on extra-alveolar vessels, but as inflation continues, resistance begins to rise as alveolar vessels are compressed.

Active Control of Pulmonary Resistance

Pulmonary vascular resistance is also subject to **active control** by **chemical and humoral substances** (Fig. 13.2). Hypoxia affects pulmonary vessels differently than it affects systemic vessels. In the systemic circulation, tissue hypoxia results in metabolic vasodilation of the arterial microcirculation. In contrast, in the pulmonary system, when **alveolar oxygen concentration** (PA_{O_2}) falls, vasoconstriction of arterioles in the hypoxic region occurs. This is the most important mechanism

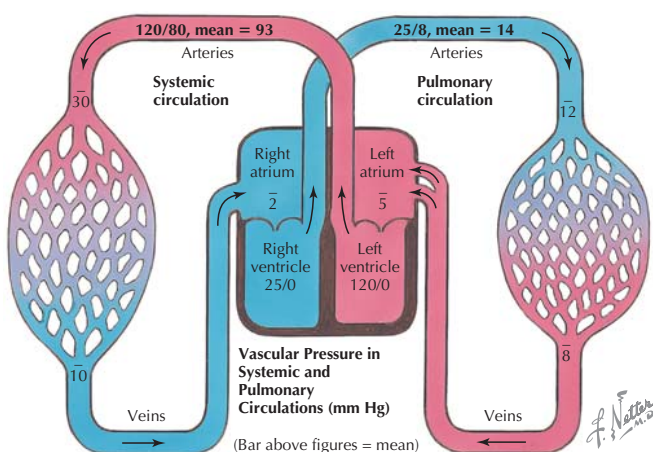
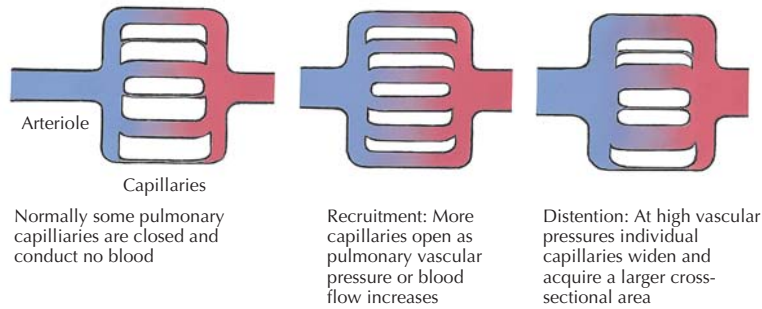
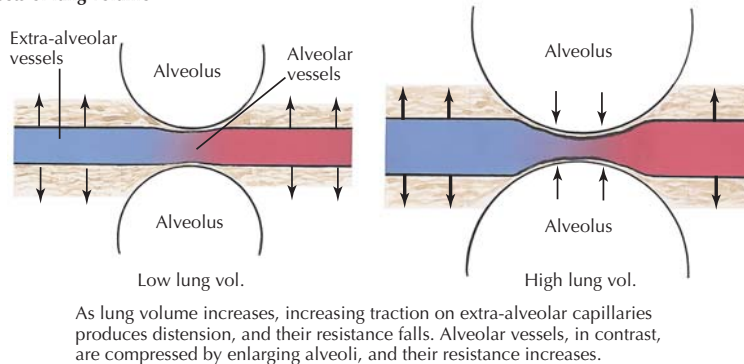


Figure 13.1 Pulmonary and Systemic Circulations The pulmonary circulation exists in series with the systemic circulation, and for this reason, equal cardiac output and stroke volumes are ejected from the right and left ventricles. Pressures and resistances are lower in the pulmonary circulation than in the systemic circulation.

A. Effects of increases in pulmonary blood flow and vascular pressures



B. Effects of lung volume



C. Effects of chemical and humoral substances

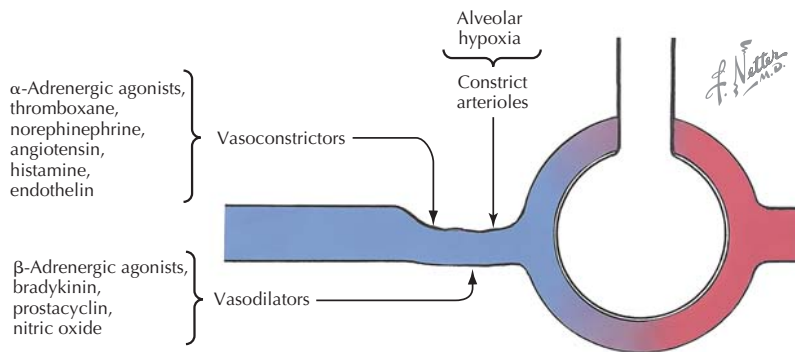


Figure 13.2 Pulmonary Vascular Resistance Resistance in the pulmonary circulation is affected by pulmonary artery pressure and blood flow, lung volume, and chemical and humoral substances. Elevation of pulmonary artery pressure causes distension of vessels and recruitment of capillaries that are otherwise collapsed (A), resulting in reduced resistance. An increase in lung volume causes traction and thus distension of extra-alveolar vessels, but results in compression of alveolar capillaries (B); the combination of these two effects results in lowest pulmonary vascular resistance at intermediate lung volumes. Pulmonary vessels also constrict and dilate in response to a number of chemical and humoral mediators (C).

for short-term regulation of regional pulmonary blood flow, and results in better perfusion of well-oxygenated portions of the lungs.

FUNCTIONAL ANATOMY OF THE LUNGS AND AIRWAYS

The human lungs consist of the three lobes of the right lung and two lobes of the left lung (Fig. 13.3). The bronchi, blood vessels, lymphatic vessels, and nerves leave and enter at the hilum of each lung. The airways consist of the trachea (wind-pipe), right and left main bronchi, smaller bronchi, and bronchioles (Figs. 13.4 and 13.5). The trachea branches into the

right and left main bronchi, which enter the lungs, further divide, and become smaller in diameter. Up to 23 generations of branching airways are present in the system beginning at the trachea and leading to the alveoli.

Conducting Zone of the Lung

Airways from the trachea to the terminal bronchioles constitute the **conducting zone** of the lungs and are not capable of gas exchange, which takes place only in the respiratory bronchioles and alveoli. The conducting zone is also known as **anatomical dead space**, due to the lack of gas exchange in this area. In an adult, the anatomical dead space contains approximately 150 milliliters (mL) of air. The trachea has

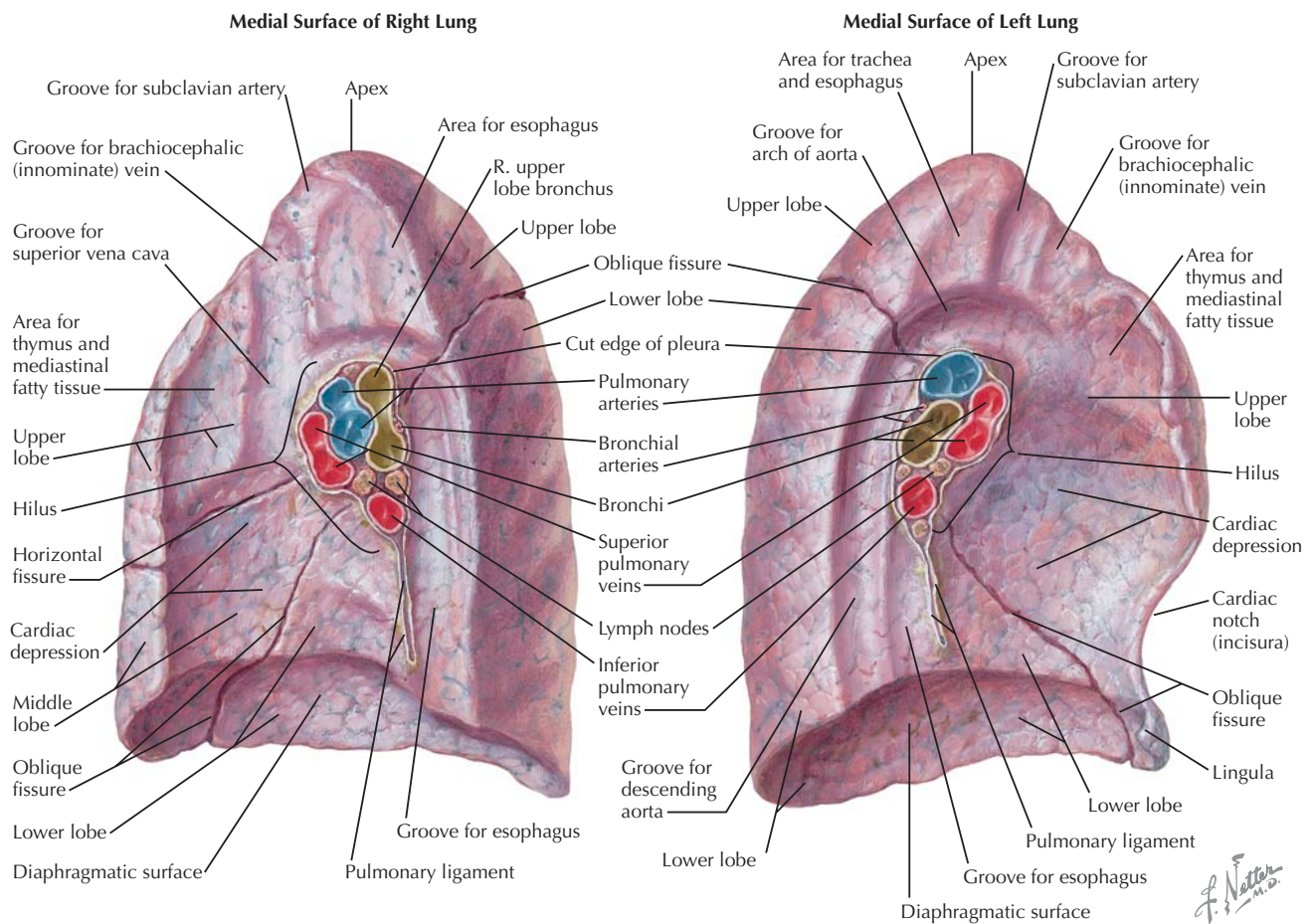


Figure 13.3 Gross Anatomy of the Lung The three lobes of the right lung and the two lobes of the left lung are depicted in this medial view of the lungs. Blood vessels, nerves, right and left main bronchi, and lymphatics enter and exit the lungs at the hilum.

cartilaginous rings that extend around three quarters of its circumference, providing the structural support necessary to hold airways open, but also allowing coughing. Cartilaginous plates are present in the walls of bronchi but are lost at the level of the bronchioles.

Most of the conducting system is lined by ciliated, **pseudostratified columnar epithelial cells**, mucous-secreting **goblet cells**, and several other cell types, with ciliated cells and goblet cells constituting the majority of cells lining the larger airways (Fig. 13.6). Mucus prevents desiccation of the epithelium and traps particulate matter in inspired air. Particles are carried upward (out of the lung) by ciliary action. This process is referred to as **mucociliary transport**. The epithelium becomes **cuboidal epithelium** in bronchioles, where ciliated cells are predominant. Goblet cells diminish and are absent in terminal bronchioles. **Clara cells** in bronchioles secrete substances that line the bronchioles and play a role in the defense system of the airways; they also act as stem cells for regeneration of bronchiolar epithelium.

The walls of the conducting airways also contain smooth muscle, which is regulated by the autonomic nervous system. Sympathetic stimulation dilates airways through the effects of norepinephrine (or circulating epinephrine) on β_2 receptors on smooth muscle cells. Parasympathetic stimulation constricts airways through muscarinic receptor activation by acetylcholine. Airway dilation is characteristic of the “fight-or-flight” response (see Chapter 7, page 87).

Respiratory Zone of the Lung

Inspired air passes through the trachea, bronchi, and bronchioles, eventually entering the terminal bronchioles, respiratory bronchioles, and alveoli. The respiratory bronchioles and alveoli are the **respiratory zone of the lungs**, where gas exchange occurs. There are an estimated 300 million alveoli in the human lungs, with an exchange area of 50 to 100 m². This enormous surface area, along with the thinness of the air–blood interface, allows for efficient gas exchange. This interface, known as the **alveolar-capillary membrane**, consists

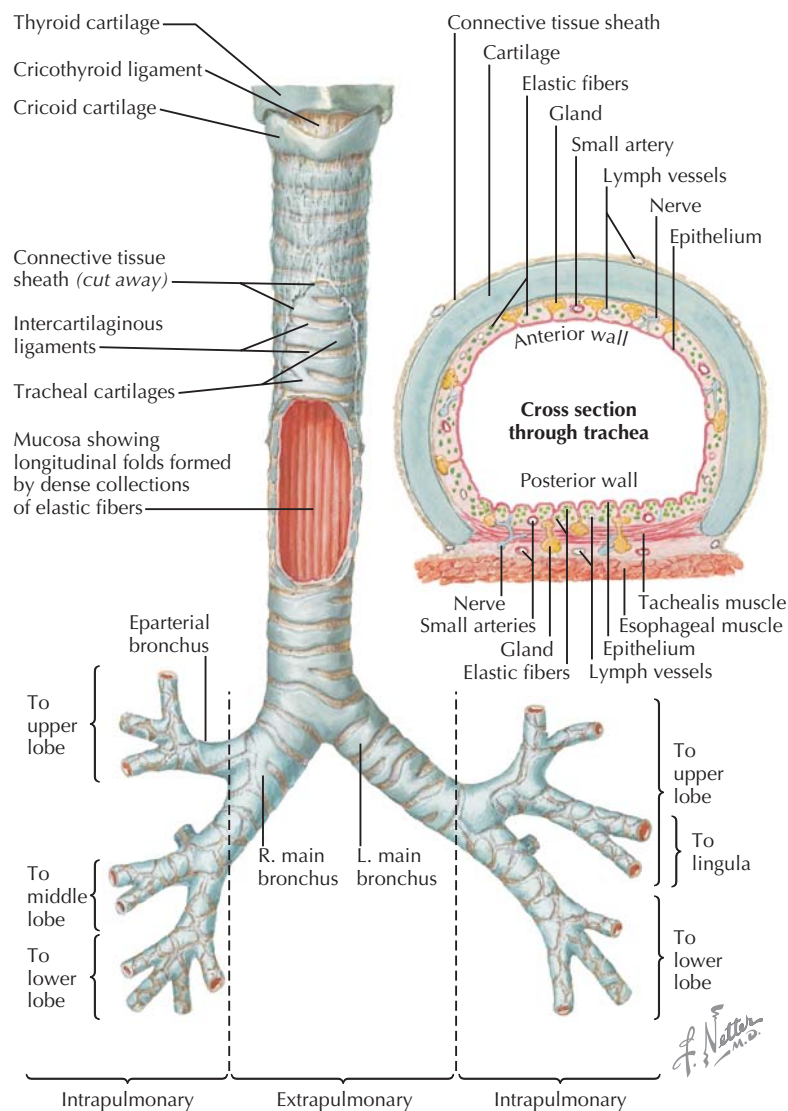


Figure 13.4 Structure of the Trachea and Major Bronchi The trachea divides into the right and left main bronchi before entering the lungs. The airways branch up to 23 times, leading to the alveoli. The airways from trachea to the terminal bronchioles are the conducting zone of the lung. With branching, diameter of airways diminishes. Collagen is less prevalent in the walls of smaller airways and is lost in the bronchioles.

of the capillary endothelium, the alveolar epithelium, and the basement membrane, upon which these two single-cell layers are formed. It is approximately 500 nanometers (nm) thick. Most of the alveolar surface is surrounded in this manner by pulmonary capillaries, allowing for “sheet flow” of blood around alveolar sacs, with a broad surface area for gas exchange (Figs. 13.7 and 13.8).

Alveoli are lined by type I and type II epithelial cells:

- **Type I epithelial cells** compose more than 90% of the surface area. Their squamous structure is adapted for gas diffusion across the alveolar-capillary membrane.

- **Type II epithelial cells** are cuboidal cells that secrete surfactant, a complex lipoprotein that lines the surface of alveoli and reduces their surface tension, increasing compliance of the lung (see “Surfactant and Surface Tension” in Chapter 14).

PULMONARY VOLUMES AND CAPACITIES

To describe pulmonary function in health and disease, it is necessary to understand the volumes and capacities associated with the lungs and breathing. There are four basic pulmonary volumes:

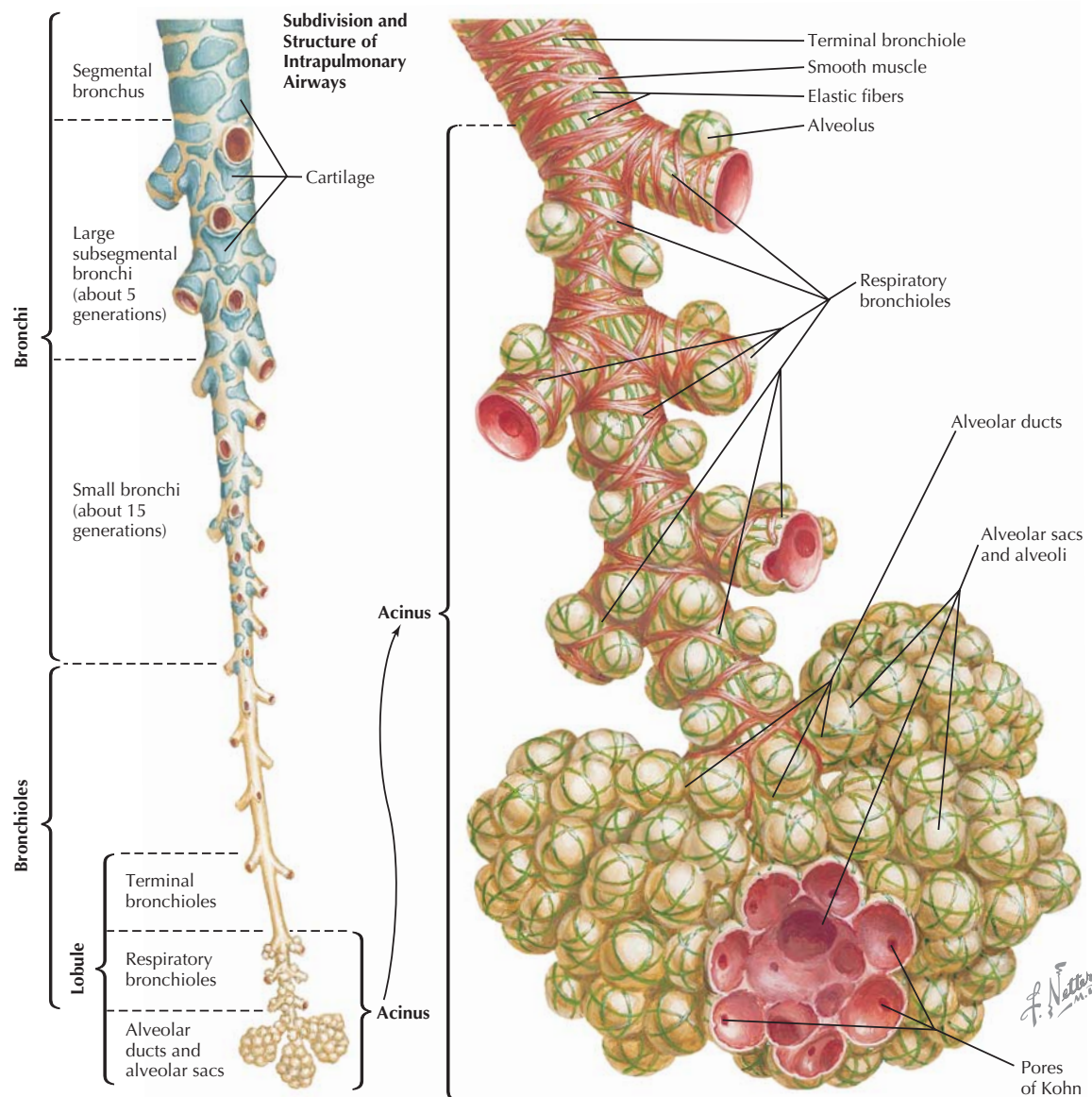
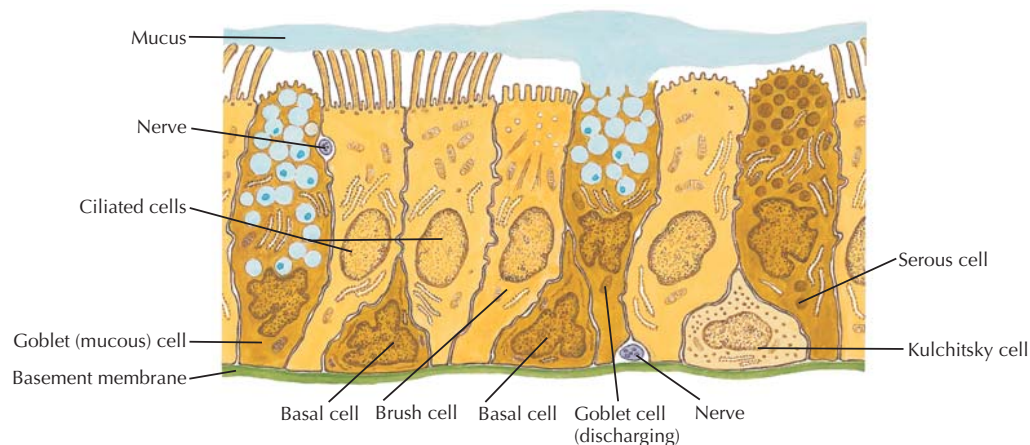


Figure 13.5 Intrapulmonary Airways After branching up to 23 times, the conducting system leads to the terminal bronchioles in the pulmonary lobules. Respiratory bronchioles and alveolar ducts give rise to alveolar sacs, the major site of gas exchange. The respiratory zone (gas exchange region) consists of structures distal to the terminal bronchioles (alveolar sacs, ducts, and respiratory bronchioles).

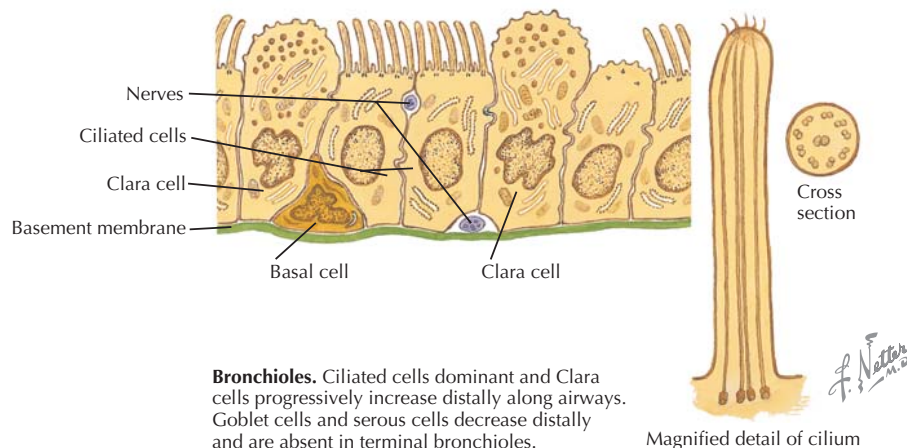
- **Tidal volume (V_T):** The volume of air inhaled and exhaled during breathing. Resting V_T is approximately 500 mL.
- **Residual volume (RV):** The volume remaining in the lungs after a maximal exhalation.
- **Expiratory reserve volume (ERV):** The additional volume a subject is *capable* of exhaling after a normal, quiet expiration.
- **Inspiratory reserve volume (IRV):** The additional volume a subject is *capable* of inhaling after a normal, quiet inspiration.

Four capacities associated with pulmonary function are:

- **Total lung capacity (TLC):** The volume of gas in the lungs after maximal inspiration. TLC is approximately 7 L in healthy adults.
- **Vital capacity (VC):** The maximum volume of air that a subject can exhale after maximal inspiration. The normal value is approximately 5 L. **Forced vital capacity (FVC)** is vital capacity measured during expiration at maximum force.



Trachea and large bronchi. Ciliated and goblet cells predominant, with some serous cells and occasional brush cells and Clara cells. Numerous basal cells and occasional Kulchitsky cells are present.



Bronchioles. Ciliated cells dominant and Clara cells progressively increase distally along airways. Goblet cells and serous cells decrease distally and are absent in terminal bronchioles.

Figure 13.6 Ultrastructure of Tracheal, Bronchial, and Bronchiolar Epithelium The trachea and bronchi are lined mainly by pseudostratified columnar epithelium and goblet cells, with several other, less common cell types. The goblet cells have the important function of mucus secretion. Kulchitsky cells are neuroendocrine-like cells that secrete paracrine factors and are part of the “diffuse neuroendocrine system” (DNES). The functions of brush and serous cells are not well defined; basal cells are pulmonary epithelial stem cells. The epithelial lining of bronchioles contains columnar epithelial cells; goblet cells are lost in terminal bronchioles. Clara cells are secretory cells of the bronchioles.

- **Functional residual capacity (FRC):** The volume remaining in the lungs after expiration during normal, quiet breathing.
- **Inspiratory capacity (IC):** The maximum volume that can be inspired after expiration during normal, quiet breathing.

Spirometry

These capacities and volumes are measured by **spirometry** and related techniques. In spirometry, the subject breathes in and out of a device known as a spirometer. Essentially, a spirometer is composed of two vessels: one contains water and



Full appreciation of pulmonary volumes and capacities requires understanding the relationships between these parameters. For example, total lung capacity is the sum of four of the described volumes:

$$TLC = RV + ERV + V_T + IRV$$

Similarly, vital capacity, functional residual capacity, and inspiratory capacity can be equated to the sums of particular lung volumes:

$$VC = IRV + V_T + ERV$$

$$FRC = ERV + RV$$

$$IC = V_T + IRV$$

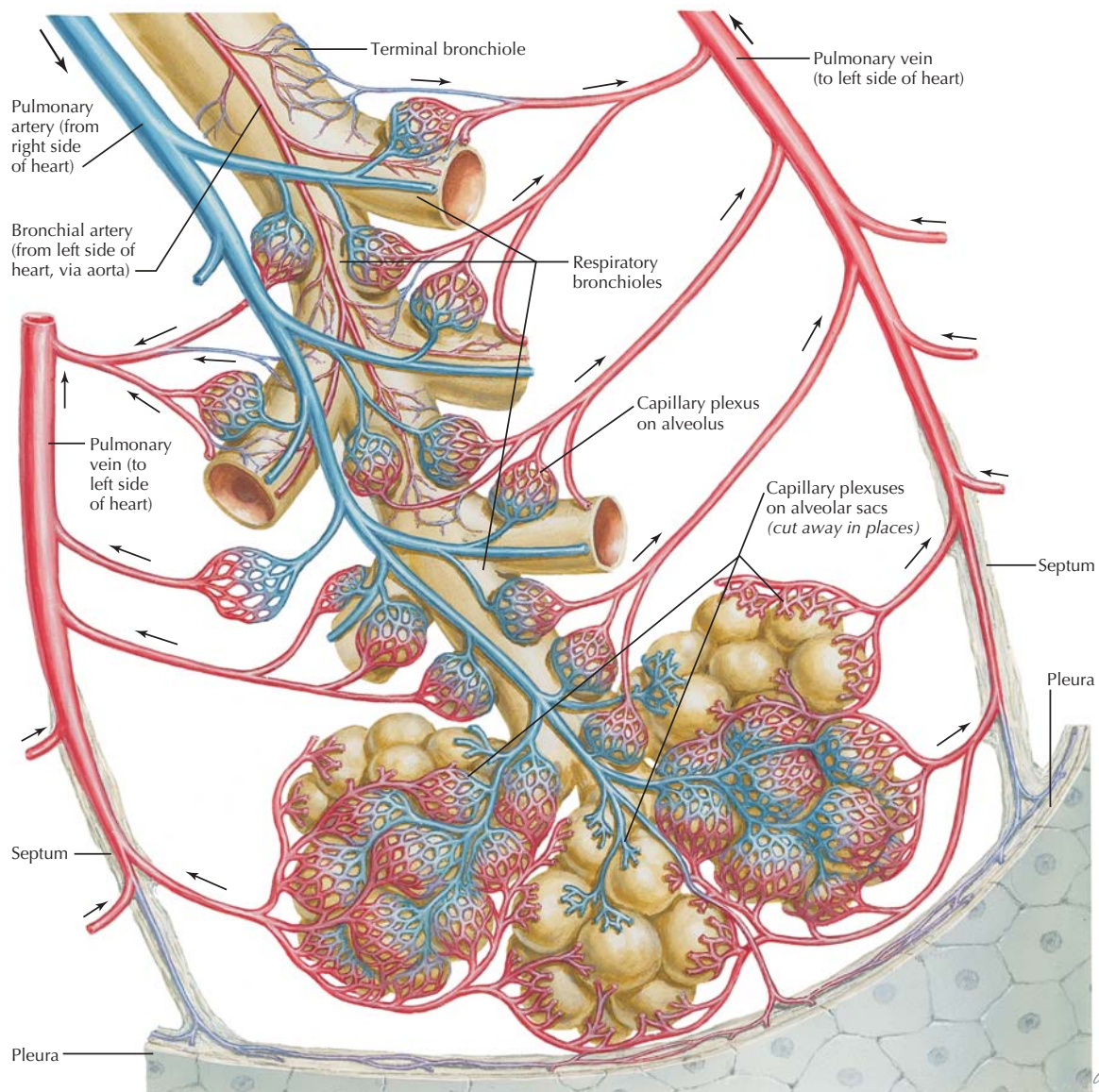


Figure 13.7 Intrapulmonary Blood Circulation The pulmonary circulation is a low-pressure, low-resistance circulation. Blood from the right ventricle and pulmonary artery is distributed to the pulmonary capillaries, where gas exchange takes place. The interface between the alveolar lumen and pulmonary capillary blood consists of a single layer of alveolar epithelium, basement membrane, and the one-cell-layer-thick capillary endothelium. Capillaries cover alveoli in this manner, providing for efficient gas exchange.

the other floats upside down in the first (Fig. 13.9). As the subject breathes through the attached tube, air flows in and out of the inner vessel, which consequently moves up and down. This up-and-down motion is recorded as a spirogram (Fig. 13.10), which is calibrated to reflect changes in volume of the inner vessel.

Tidal volume is measured by spirometry during normal, quiet breathing and is the difference between end inspiratory and end expiratory levels (see Fig. 13.10). Note that the actual

end inspiratory and end expiratory volumes are unknown, because the spirometer records *changes* in volume rather than actual volume inside the lung. By having the subject expire maximally, expiratory reserve volume can be determined, because ERV is the difference between the resting end expiratory level and maximal expiratory level. Likewise, inspiratory reserve volume can be measured by having the subject inspire maximally and comparing the maximal inspiratory level to the end inspiratory level during normal, quiet breathing. Vital capacity (VC) and inspiratory capacity (IC) can be

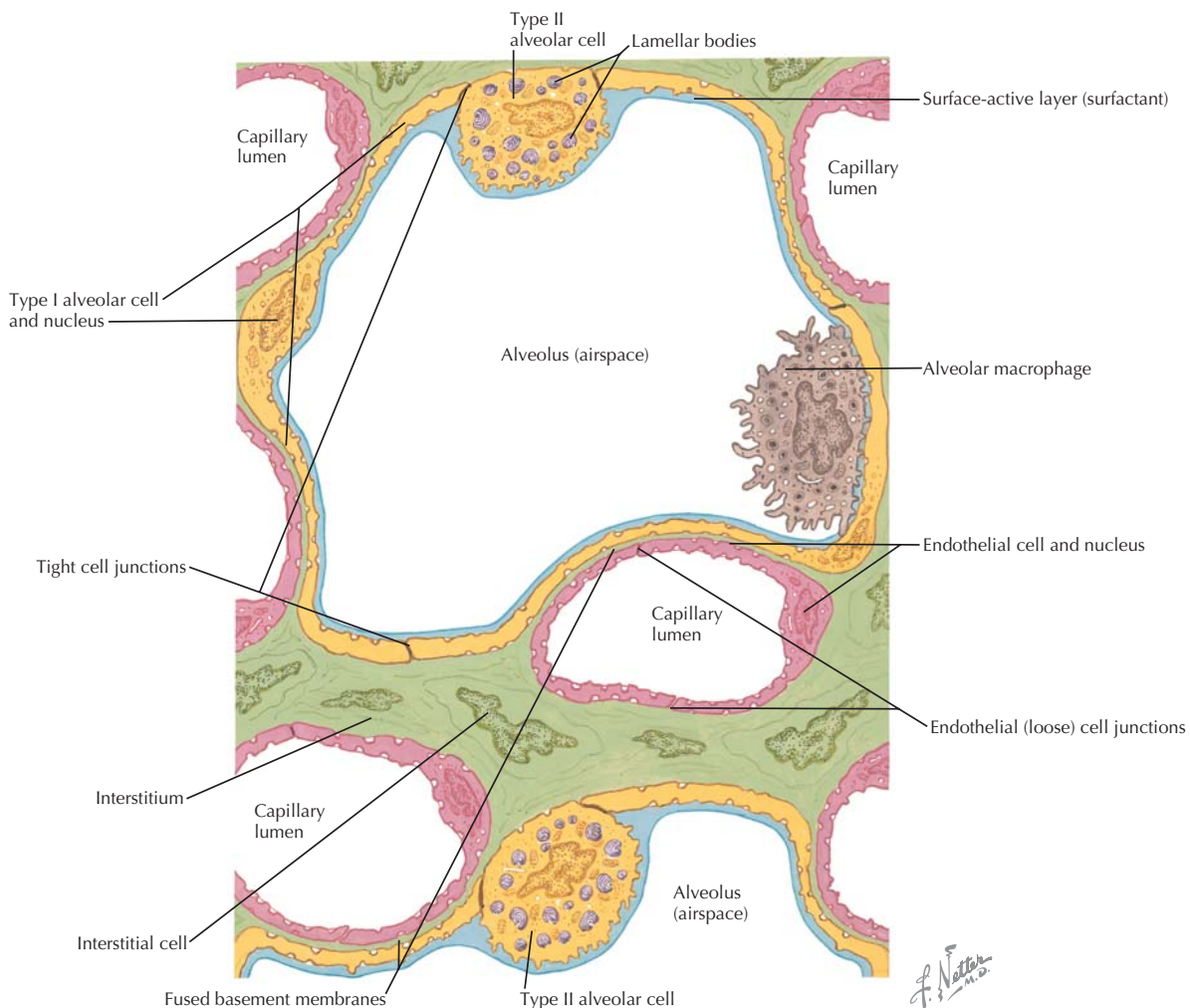


Figure 13.8 Ultrastructure of Pulmonary Alveoli and Capillaries The alveolar epithelium consists of type I and type II alveolar epithelial cells. Type I cells constitute the largest surface area; type II epithelial cells secrete surfactant. Gas diffusion takes place across the thin alveolar-capillary membrane composed of alveolar epithelium, basement membrane, and capillary endothelium. Interstitial tissue is minimal in most of the interface between the alveolar epithelium and capillary endothelium in healthy lungs.

determined by similar comparisons: VC is the difference between the maximal inspiratory and expiratory levels; IC is the difference between maximal inspiratory level and resting end-expiratory level.

Measurement of Capacities (FRC, RV, and TLC)

In order to measure total lung capacity, residual volume, and functional residual capacity, one of these parameters must be measured indirectly, for example by nitrogen washout, helium dilution, or body plethysmography. In the **helium dilution technique**, a small volume of helium is added to the spirometer before the subject begins the test, resulting in a known initial helium concentration within the volume of the system. When the subject begins breathing (beginning at FRC), this

initial helium concentration becomes diluted, as the gas equilibrates between the lungs of the subject and the spirometer. Because helium does not diffuse through the alveolar-capillary membrane and is inert, a stable equilibrium is reached quickly. This final concentration is dependent only on the initial concentration of helium within the spirometer and the volume of the spirometry system plus FRC. After measuring the final concentration of helium, FRC can be determined from the formula:

$$C_1 V_s = C_2 \times (V_s + \text{FRC})$$

where C_1 is the initial, measured concentration of helium in the spirometer, V_s is the known volume of the spirometry system, and C_2 is the measured concentration of helium during breathing (after equilibrium is reached).

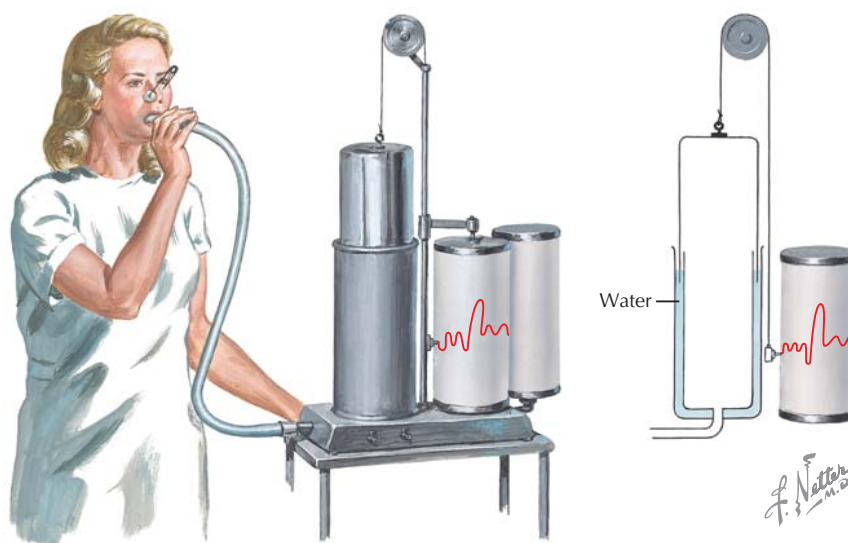


Figure 13.9 The Spirometer Pulmonary function testing involves multiple instruments and techniques. One basic instrument is the spirometer. The subject breathes in and out of a tube, resulting in vertical displacement of the inner canister, which floats upside down in water. The recorded tracing is analyzed to obtain several basic lung volumes and capacities (see Fig. 13.10).

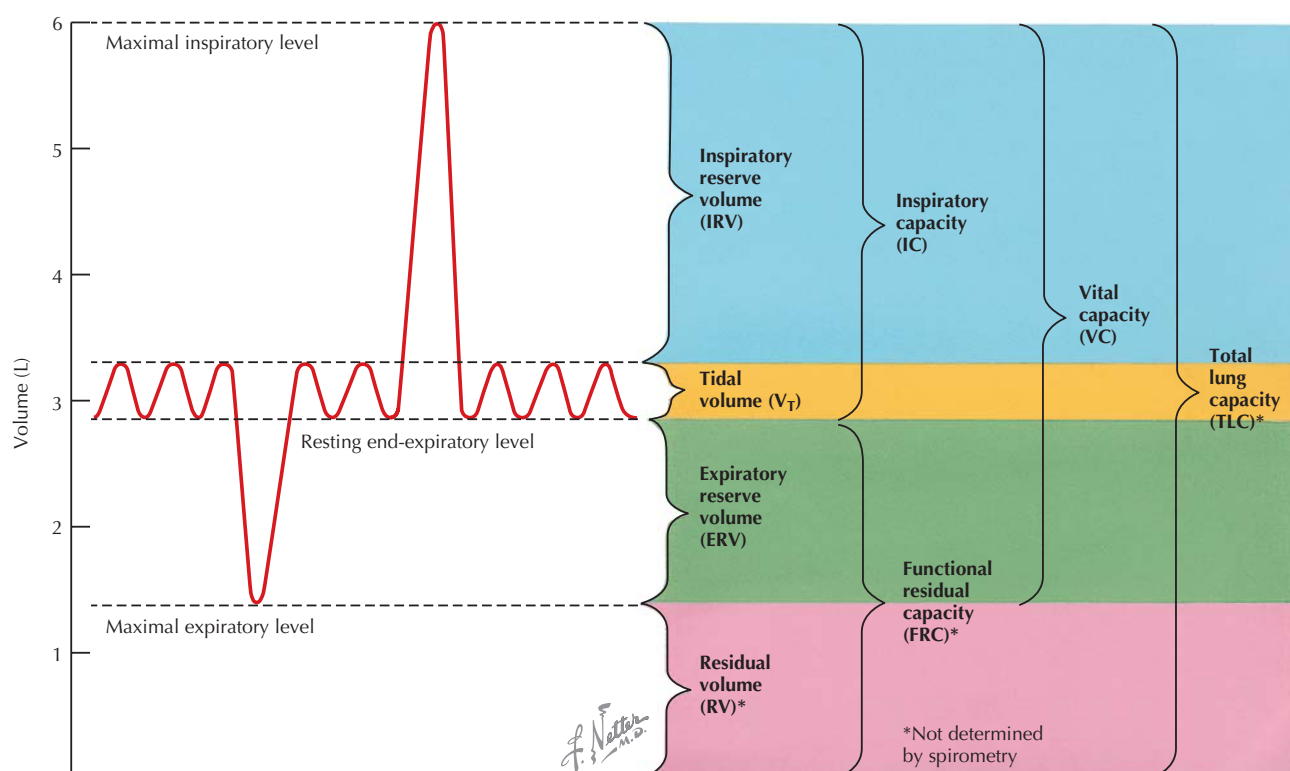


Figure 13.10 Measurement of Lung Volumes and Capacities by Spirometry In this spirogram, beginning on the left, the subject breathed quietly for a few breaths, exhaled maximally, breathed quietly for a few breaths, inhaled maximally, and then breathed normally again. Note that residual volume, functional residual capacity, and total lung capacity cannot be measured by spirometry alone. FRC is measured by another technique (often helium dilution); when FRC is known, TLC and RV can be calculated from the spirometry tracing. The volumes represented in the tracing are those of a typical healthy adult.

Once FRC is known, TLC and RV can be readily calculated from spirometry measurements (see Fig. 13.10):

$$\begin{aligned}\text{TLC} &= \text{FRC} + \text{IC} \\ \text{RV} &= \text{TLC} - \text{VC}\end{aligned}$$

Whole-body plethysmography is based on **Boyle's law**, which states that the product of pressure and volume for a gas is constant:

$$P_1 V_1 = P_2 V_2$$

To employ whole-body plethysmography to measure FRC, the subject is put into an airtight box equipped with a mouthpiece through which he breathes outside air. During normal, quiet breathing, the subject is asked to stop and relax after a normal, quiet exhalation. At that point, the mouthpiece is closed, and the subject attempts to inhale through the closed mouthpiece. This attempt causes the air in the lungs (FRC) to expand due to the negative pressures created by the expansion of the chest wall, and results in an equal reduction of the volume of the air in the box outside the patient's body. The pressure of the air in the box increases due to this reduction in volume. Changes in pressure in the box are used to calculate the change in volume in the box using Boyle's law. This change in volume is the same as the change in volume of the lungs from the resting value (FRC). Based on the change in pressure in the respiratory system (measured at the mouthpiece during the attempt to inhale) and the change in volume of the lungs, the initial volume (FRC) can be calculated using Boyle's law.

VENTILATION AND ALVEOLAR GAS COMPOSITION

Ventilation is the movement of air in and out of the respiratory system. During normal, quiet breathing, the volume of air inhaled and exhaled with each breath is approximately 500 mL. This is known as the resting **tidal volume**. The normal, resting **respiratory rate** is 12 to 20 breaths per minute (min). Taking an average rate of 15 breaths per minute, the **minute ventilation** ($\dot{V}_E$, the volume exhaled per minute) is calculated by the formula:

$$\dot{V}_E = R \times V_T$$

where R is the respiratory rate and V_T is the tidal volume. Thus, 15 breaths/min $\times$ 500 mL yields a $\dot{V}_E$ of 7500 mL/min (7.5 L/min).

Minute ventilation ($\dot{V}_E$) is greater than **alveolar ventilation** ($\dot{V}_A$, ventilation of the respiratory zone of the lungs), because a portion of the tidal volume remains in the anatomical dead space (the conducting zone) and is not involved in gas exchange. Because anatomical dead space is approximately 150 mL, this **dead space ventilation** ($\dot{V}_D$) is roughly 150 mL $\times$ 15/min, or 2250 mL/min. Alveolar ventilation is calculated by the formula:



The symbols used in respiratory physiology can be confusing. The uppercase V is used to designate volume, whereas airflow rates are designated by the symbol $\dot{V}$. Uppercase A used as a subscript, as in $\dot{V}_A$, specifies that the parameter refers to the alveolar space. Thus, $\dot{V}_A$ is alveolar ventilation. Lowercase a and v are used to specify arterial and venous measurements, respectively, as in $P_{a\text{CO}_2}$ or $P_{v\text{CO}_2}$ (partial pressure of carbon dioxide in arterial and venous blood, respectively). $P_{a\text{CO}_2}$ refers to partial pressure of carbon dioxide in alveolar gas.

$$\dot{V}_A = R(V_T - V_D)$$

where R is respiratory rate, V_T is tidal volume, and V_D is dead space volume. Stated another way,

$$\dot{V}_A = \dot{V}_E - \dot{V}_D$$

Of the 500 mL tidal volume, only 350 mL is entering the alveoli with each breath, and of the 7500 mL/min minute ventilation, only the alveolar ventilation of 5250 mL/min (7500 mL/min $-$ 2250 mL/min) is available for gas exchange.

Composition of Alveolar Air

The composition of alveolar air is dependent on several factors, including the composition of inspired air, alveolar ventilation, and the concentration of dissolved gases in mixed venous blood. Our atmosphere is composed of 21% oxygen, 79% nitrogen, and less than 1% other gases, including carbon dioxide, with a total atmospheric pressure of 760 mm Hg at sea level. Gas concentrations may also be expressed as fractional concentrations in an inspired gas mixture; in our atmosphere, $\text{FiO}_2 = 0.21$ and $\text{FiN}_2 = 0.79$. According to **Dalton's law**, the sum of the partial pressures of gases in a mixture is equal to the total pressure. Thus, for dry air at sea level:

$$\begin{aligned}P_{\text{O}_2} &= 0.21 \times 760 \text{ mm Hg} = 160 \text{ mm Hg} \\ P_{\text{N}_2} &= 0.79 \times 760 \text{ mm Hg} = 600 \text{ mm Hg} \\ P_{\text{tot}} &= 760 \text{ mm Hg}\end{aligned}$$

As air is inspired, it warms rapidly to body temperature and becomes saturated with water vapor. At 37°C, the vapor pres-



At the same minute ventilation ($\dot{V}_E$), deep, slow breathing yields greater alveolar ventilation ($\dot{V}_A$) than rapid, shallow breathing. Comparing alveolar ventilation when respiratory rate is 15/min and tidal volume is 500 mL to alveolar ventilation at a respiratory rate of 30/min and tidal volume of 250 mL:

$$\begin{aligned}\dot{V}_A &= 15/\text{min} (500 \text{ mL} - 150 \text{ mL}) = 5250 \text{ mL/min} \\ \dot{V}_A &= 30/\text{min} (250 \text{ mL} - 150 \text{ mL}) = 3000 \text{ mL/min}\end{aligned}$$

Thus, slower, deeper ventilation produces greater alveolar ventilation than more rapid ventilation at proportionally less tidal volume.

sure of water is 47 mm Hg and must be accounted for when determining the gas composition of inspired air. In **inspired air**:

$$\begin{aligned}P_{H_2O} &= 47 \text{ mm Hg} \\P_{O_2} &= 0.21 \times (760 - 47) \text{ mm Hg} = 150 \text{ mm Hg} \\P_{N_2} &= 0.79 \times (760 - 47) \text{ mm Hg} = 563 \text{ mm Hg} \\P_{\text{tot}} &= (150 + 563 + 47) \text{ mm Hg} = 760 \text{ mm Hg}\end{aligned}$$

This composition of inspired air is constant throughout the conducting zone of the lung, where no gas exchange occurs. Within the respiratory zone, oxygen diffuses from alveolar air to blood, while carbon dioxide diffuses from blood to alveolar air, resulting in alveolar air composition that is different from composition of inspired air. The relationship between partial pressure of oxygen and carbon dioxide in alveolar air is described by the important **alveolar gas equation**:

$$P_{A_{O_2}} = P_{I_{O_2}} - P_{A_{CO_2}}/R$$

where $P_{A_{O_2}}$ and $P_{I_{O_2}}$ are the partial pressures of oxygen in alveolar air and inspired air, respectively; $P_{A_{CO_2}}$ is partial pressure of carbon dioxide in alveolar air; and R is the **respiratory quotient**, which usually has a value of 0.8. This equation can be used to predict the partial pressure of oxygen in alveolar air based on measurement of carbon dioxide in systemic arterial blood in a healthy person, because P_{CO_2} is normally fully equilibrated between blood and alveolar air in the alveolar capillaries. For example, at 760 mm Hg atmospheric pressure (sea level), when $P_{a_{O_2}}$ in an arterial blood gas determination (ABG) is 40 mm Hg:

$$P_{A_{O_2}} = 150 \text{ mm Hg} - 40 \text{ mm Hg}/0.8 = 100 \text{ mm Hg}$$

Because PO_2 is also fully equilibrated between alveolar air and blood as it passes through alveolar capillaries, and nearly all of pulmonary blood flow is exposed to alveolar air, the normal value for $P_{a_{O_2}}$ is nearly 100 mm Hg. Partial pressures of oxygen and carbon dioxide in inspired and alveolar air and mixed venous and arterial blood are illustrated in Figure 13.11.

DIFFUSION OF GASES

Diffusion of gas ($\dot{V}_{\text{gas}}$) between alveolar gas and capillary blood follows Fick's law,

$$\dot{V}_{\text{gas}} = \frac{A \times D(P_1 - P_2)}{T}$$

where A is the area of a membrane separating two compartments, T is the thickness of the membrane, D is the diffusion constant, and P_1 and P_2 are the gas concentrations in the two compartments. Thus, diffusion of a gas is:

- Directly related to the surface area for diffusion.
- Directly related to the difference in partial pressure of the gas on each side of the membrane.
- Directly related to the diffusion constant of the gas.
- Inversely related to the thickness of the membrane.

The diffusion constant of a gas is directly related to solubility and inversely related to the square root of its molecular weight.

Perfusion-Limited Gas Transport

The concentrations of dissolved gases in blood equilibrate with the gas concentrations of alveolar air as blood flows through pulmonary capillaries (Fig. 13.12). Under resting conditions, the transit time for blood through the alveolar capillary is only approximately 0.75 seconds (sec). In normal, healthy lungs at rest, oxygen and carbon dioxide are equilibrated between blood and alveolar gas by the time blood has passed one third of the way through an alveolar capillary. Transport of these gases is **perfusion-limited** under these conditions, because the only way to increase gas transfer is to increase perfusion (blood flow, rather than diffusion, is the limiting factor for exchange). As a consequence, cardiac output can be increased substantially without compromising oxygenation of blood despite reduced transit time through the alveolar capillaries, for example during exercise. Nitrous oxide (laughing gas; N_2O) is a textbook example of perfusion-limitation. N_2O , unlike many other gases including O_2 and CO_2 , is not bound by blood components but exists only as dissolved gas in blood. As a result, in a subject breathing a gas mixture containing N_2O , blood and alveolar P_{N_2O} equilibrate rapidly in blood flowing through the alveolar capillary, reaching this equilibrium one fifth of the way through the capillary.



The respiratory quotient (R) is the ratio of carbon dioxide production to oxygen consumption and is dependent on metabolism. For pure carbohydrate metabolism, R has a value of 1.0, because oxidative metabolism of one mole of glucose requires six moles of oxygen and produces six moles of carbon dioxide. During pure lipid metabolism, the approximate value of R would be 0.7, whereas the respiratory quotient for amino acids varies. Under metabolic conditions involving oxidation of a typical mixture of substrates, the respiratory quotient is approximately 0.8 (0.8 moles of carbon dioxide are produced for each mole of oxygen consumed).



Transit time is the time required for blood or a formed element to pass through a portion of the circulation. Because blood volume in the entire pulmonary circulation is approximately 500 mL and cardiac output at rest is about 5000 L/min (83 ml/sec), transit time through the entire pulmonary circulation is approximately 6 seconds. Of this, about 0.75 second is transit time through the alveolar capillaries, during which all exchange of gases takes place. In healthy lungs, oxygen is equilibrated between blood and alveolar gas by the time blood has passed one third of the way through an alveolar capillary.

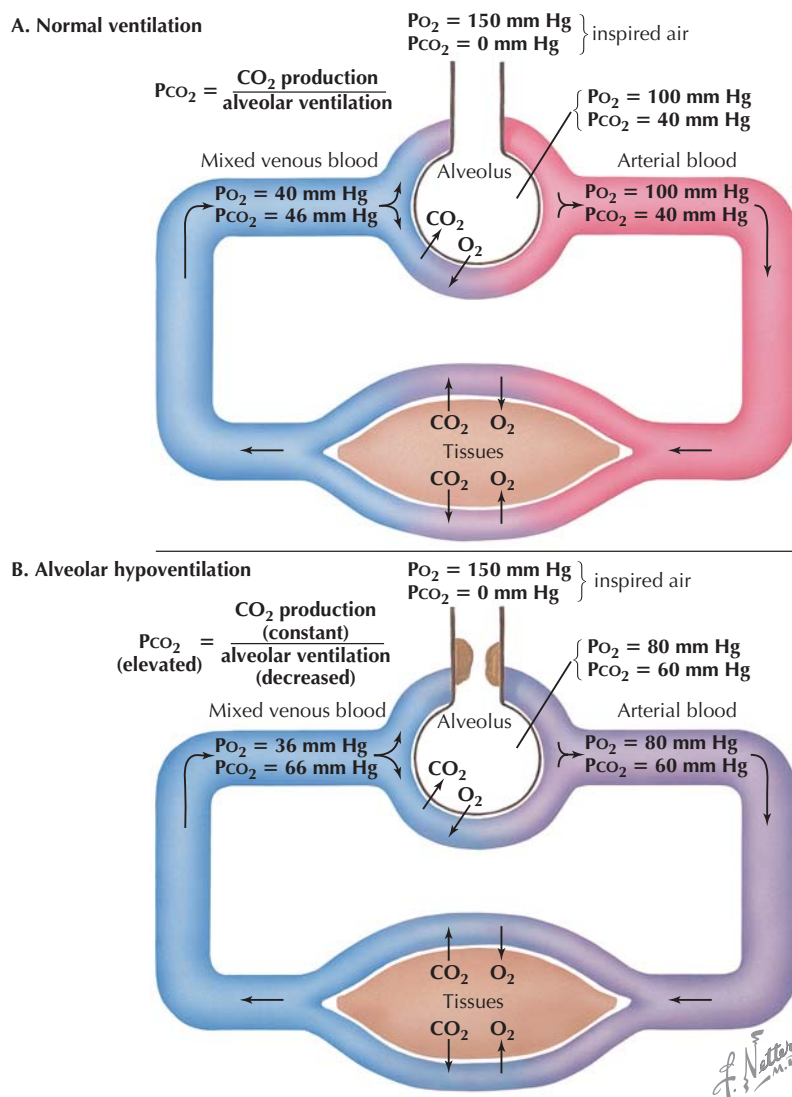


Figure 13.11 Partial Pressures of Oxygen and Carbon Dioxide in Blood and Alveolar Air
 The normal partial pressures of gases in mixed venous blood entering the alveolar capillaries are indicated on the left side of part A. As blood flows through the alveolar capillaries in healthy lungs, oxygen and carbon dioxide levels equilibrate between alveolar air and blood. Thus, partial pressures of oxygen and carbon dioxide determined by arterial blood gas (ABG) measurement are approximately equal to the partial pressures in alveolar air in healthy subjects. During hypoventilation (B, indicated by partial blockage of the airway), P_{O_2} is reduced and P_{CO_2} is elevated in arterial blood (the subject is hypoxic and hypercapnic) as well as in the alveolar gas.

Diffusion-Limited Gas Transport

During very strenuous aerobic exercise in which cardiac output is greatly increased, or in some disease states such as interstitial fibrosis, oxygen transport may be **diffusion-limited**. In other words, P_{O_2} is not completely equilibrated between alveolar air and blood leaving the alveolar capillary. The classic example of a diffusion-limited gas is carbon monoxide (CO). When a subject breathes a gas mixture containing CO, CO diffuses from the alveolar air to blood. However, CO is bound to hemoglobin with very high affinity, and as a result, large

amounts of CO can be transferred to blood with little change in blood P_{CO} . Thus, the partial pressures of CO in alveolar air and blood do not fully equilibrate during the transit of blood through the alveolar capillary, and its transport is limited only by the diffusion capacity of the alveolar membrane.

Diffusion in the lung can be assessed using a measurement known as **diffusion capacity of the lung for CO (DLCO)**. CO is useful for this purpose because its transport is diffusion-limited. According to Fick's law,

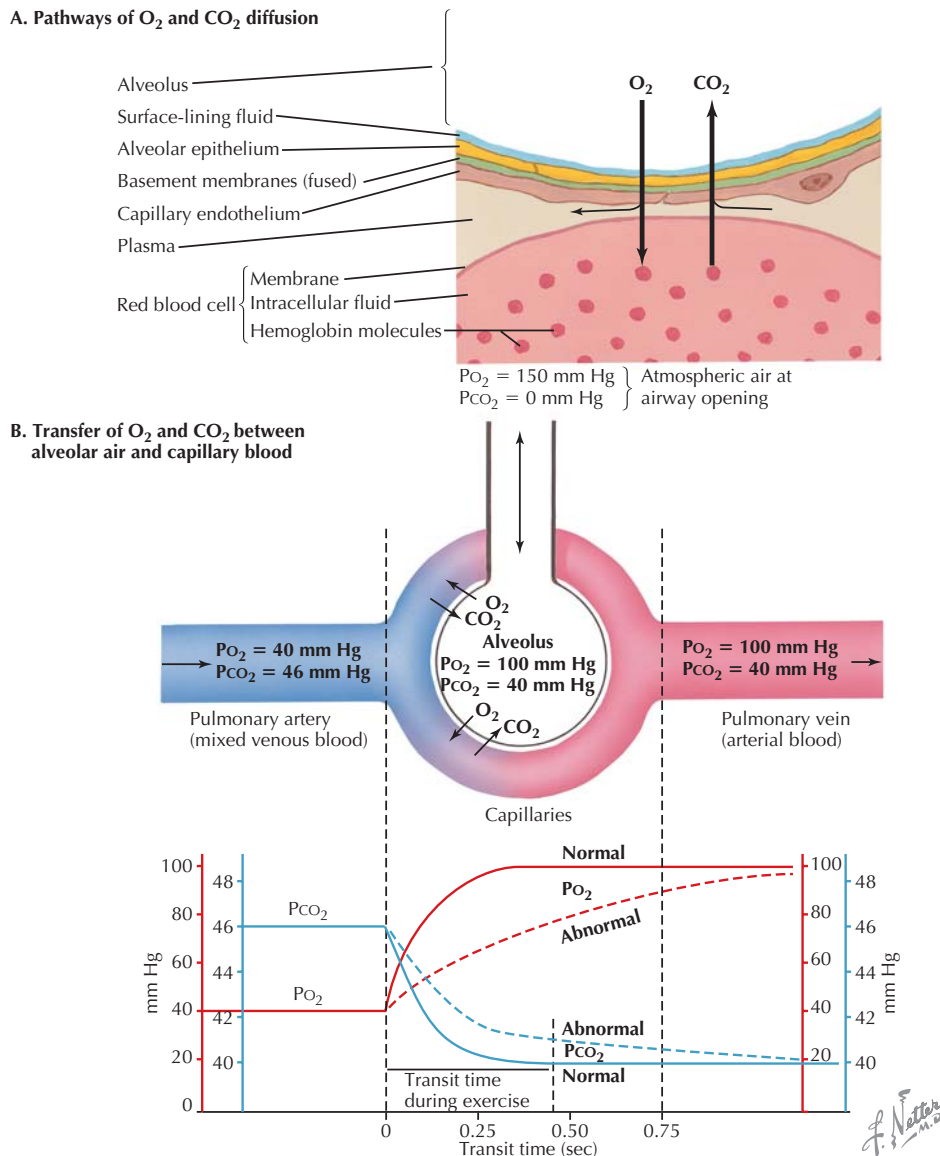


Figure 13.12 O₂ and CO₂ Exchange Diffusion of carbon dioxide and oxygen between blood and alveolar air occurs by diffusion through the thin alveolar-capillary membrane (A). Partial pressures of gases in inspired air, alveolar air, and blood are illustrated in the center panel. At resting cardiac output, blood normally traverses the length of the alveolar capillary in 0.75 sec, and O₂ and CO₂ are equilibrated between blood and alveolar air as blood passes through the first third of the capillary (B). When the alveolar-capillary membrane is thickened by interstitial fibrosis, the resulting diffusion barrier impedes gas exchange (*dashed lines*).

$$\dot{V}_{\text{CO}} = \frac{A \times \text{DCO} \times (\text{PCO}_1 - \text{PCO}_2)}{T}$$

$$\text{DLCO} = \frac{\dot{V}_{\text{CO}}}{\text{P}_{\text{ACO}}}$$

DLCO is defined as a transfer factor equal to $(A/T) \times \text{DCO}$ and is substituted in the above equation, yielding

$$\dot{V}_{\text{CO}} = \text{DLCO} \times (\text{PCO}_1 - \text{PCO}_2)$$

Rearranging and substituting P_{ACO} for $\text{PCO}_1 - \text{PCO}_2$ (because the concentration of CO in blood is initially zero),

VENTILATION AND PERFUSION GRADIENTS

Optimal function of the lung requires proper balance between ventilation and perfusion. In other words, ventilation should be matched to blood flow for efficient exchange of gases

CLINICAL CORRELATE

DLCO and Interstitial Pulmonary Fibrosis

Interstitial fibrosis is a type of restrictive lung disease, in which functional lung volume is reduced. It is an inflammatory disease in which deposition of connective tissue and subsequent scar formation in the interstitial space of the alveolar-capillary membrane results in thickening of the membrane. As a consequence, diffusion across the alveolar-capillary membrane is impeded, and mechanical properties of the lung are altered (resulting in “restriction”; discussed in context in Chapter 14). Changes in the diffusing capacity of the lung in interstitial fibrosis are typically assessed by measurement of DLCO. Beginning at residual volume, the subject takes in a single, large breath of air containing 0.3% CO and 10% helium, inhaling to total lung capacity, holds this breath for 10 seconds, and then exhales. A sample of exhaled air is collected after exhalation of dead space volume for measurement of the final concentration of gases in alveolar air. DLCO is then determined by the formula $DLCO = \dot{V}_{CO}/P_{A_{CO}}$. $P_{A_{CO}}$ in this formula is the initial value upon inspiration of the gas mixture and is calculated based on CO concentration in inspired air and helium dilution; $\dot{V}_{CO}$ is calculated based on alveolar volume, breath-hold time, and the change in CO concentration in alveolar air.

between the environment and the blood. However, neither ventilation nor perfusion of the lung is uniform from apex to base.

In the standing position, the weight of the lungs stretches the top portion of the organ, such that the alveoli near the apex have larger volume than those near the base. Thus, the smaller alveoli toward the base of the lung have greater compliance than those that are already stretched by gravitational forces, and as a result, ventilation is greater toward the bottom of the lung. In other words, there is an increasing ventilation gradient from top to bottom of the lung.

Zones of the Lung

An even larger gradient (than the one for ventilation) exists for pulmonary blood flow (perfusion) in the standing position (Fig. 13.13), whereby greater flow occurs toward the bottom of the lungs. This is the result of gravitational effects on vascular pressures and the relationship between vascular pressure and alveolar pressure as blood flows through alveolar capillaries. Perfusion pressure in most regional circulations is the difference between arterial and venous pressure. However,

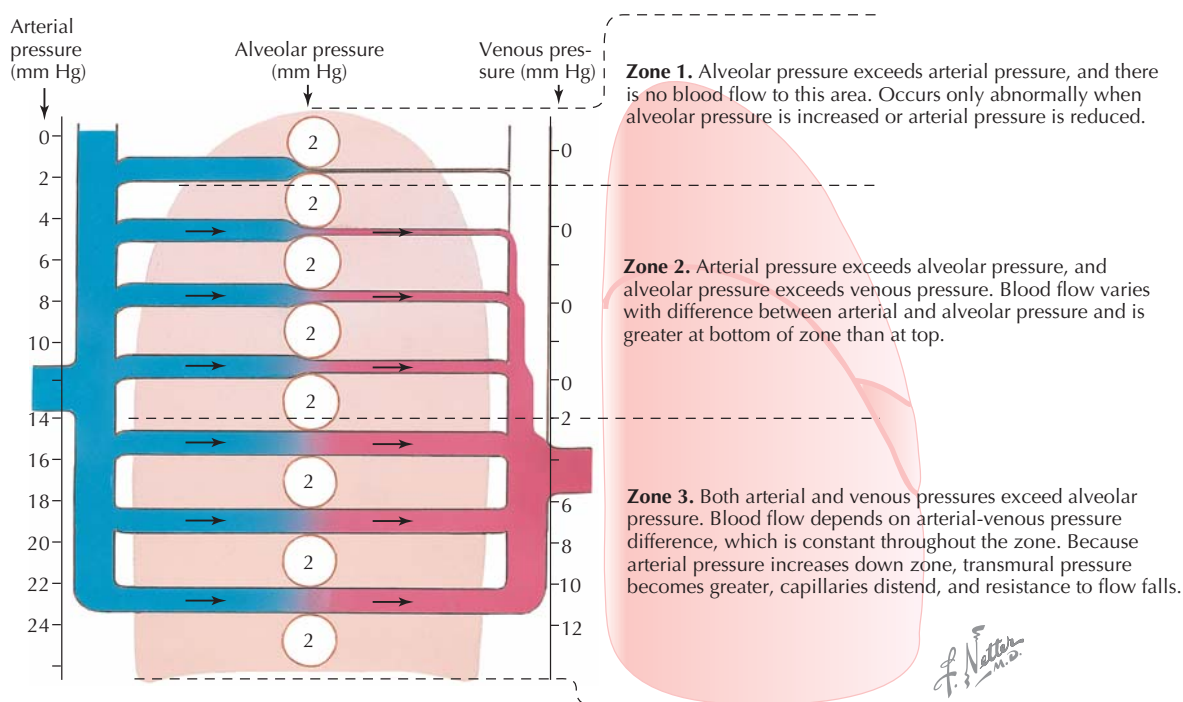


Figure 13.13 Distribution of Pulmonary Blood Flow As a result of gravity, blood flow through the lungs is not uniformly distributed in the standing position. At the apex of the lungs, hydrostatic pressure is reduced, potentially resulting in collapse of capillaries by alveolar pressure; as a result, flow is reduced toward the apex, and may cease if alveolar pressure is increased or blood pressure is reduced (zone 1). In the middle of the lungs (zone 2), arterial pressure exceeds alveolar pressure, but alveolar pressure is higher than venous pressure. Flow in zone 2 is more than that in zone 1, but less than flow in zone 3, where higher hydrostatic pressure causes distension of vessels and therefore reduced vascular resistance.

in the pulmonary system, as blood flows through a capillary, the rate of flow is potentially affected by air pressure in the alveoli on either side of the capillary. Blood enters the lung at the hilum, approximately midway between apex and base; above this point of entry, hydrostatic pressure in the pulmonary vessels is reduced by the effect of gravity. At the very top of the lung, alveolar pressure may even exceed arterial pressure under some conditions, producing a region of no blood flow, **zone 1**. In regions of the lung in which alveolar pressure is higher than venous pressure but less than arterial pressure, perfusion pressure is equal to arterial pressure minus alveolar pressure. This occurs in **zone 2** of the lung (see Fig. 13.13). Thus, as blood passes through alveolar capillaries in zone 2, the pressure gradient for blood flow is the difference between pulmonary arterial pressure and the pressure within the alveoli, which exceeds venous pressure. In contrast, in **zone 3** (lower portions of the lung) pulmonary vascular pressures are elevated by the effect of gravity, and both arterial and venous pressures exceed alveolar pressure. The higher hydrostatic pressure causes distension of vessels and therefore reduced resistance; as a result, blood flow is greatest in zone 3 of the lung.

The Ventilation-to-Perfusion Ratio

As a result of these gradients in ventilation and perfusion, the ratio $\dot{V}_A/\dot{Q}_C$ (where $\dot{Q}_C$ is pulmonary capillary blood flow) is greatest at the top of the lung and lowest at the bottom (Fig. 13.14), and ventilation and perfusion are best matched in the middle portion. Because resting pulmonary blood flow is approximately 5 L/min (equal to cardiac output of the systemic circulation) and alveolar ventilation is also approximately 5 L/min, $\dot{V}_A/\dot{Q}_C$ is approximately 1 in this middle region, less than 1 at the base of the lung, and more than 1 toward the apex.

Dead Space and Shunt: Extremes of $\dot{V}_A/\dot{Q}_C$ Imbalance

Dead space and **shunt flow** are extremes of imbalance. Alveoli that are ventilated but not perfused constitute **physiological dead space** (as opposed to anatomical dead space). In dead space, $\dot{V}_A/\dot{Q}_C = \infty$. In contrast to dead space (ventilation without perfusion), **shunt flow** represents perfusion without ventilation. In other words, shunted blood returning to the left heart has not been exposed to ventilated alveoli. If alveolar capillaries are perfused in an area of the lung that is not ventilated (for example, when an airway is blocked) **physiological shunt** is present ($\dot{V}_A/\dot{Q}_C = 0$). Shunt can also be **anatomical** (bypassing the alveoli). In addition to blood flow originating in the pulmonary arteries, the pulmonary veins receive some flow from the bronchopulmonary (bronchial) circulation, which originates from the systemic circulation and perfuses tissues in the conducting zone of the lungs. Thus, blood from the bronchopulmonary circulation returning to the left heart is not oxygenated. The venous admixture of oxygenated and deoxygenated blood resulting from anatomical shunt accounts for most of the small **alveolar-to-systemic arterial PO_2 gradient (A-a PO_2 gradient)** in healthy subjects (6 to 9 mm Hg).

While $\dot{V}_A/\dot{Q}_C$ varies even within normal lungs, ventilation-perfusion imbalances are a central issue in pulmonary pathophysiology (discussed in various contexts in following pages).

HYPOXEMIA

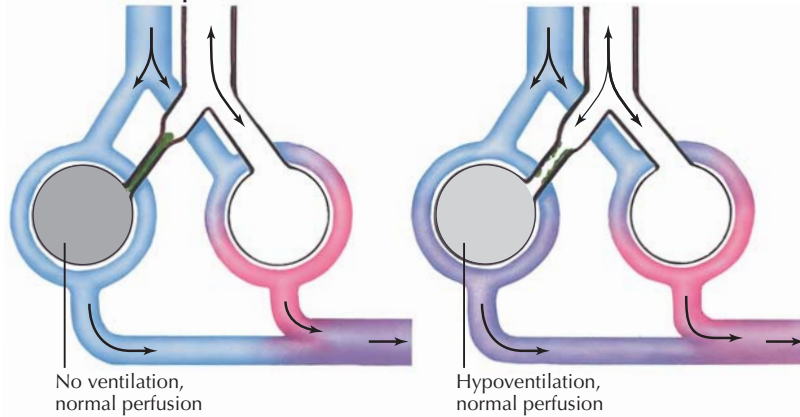
Hypoxemia (low Pa_{O_2}) may result from one of five causes, all of which reflect problems related to ventilation, perfusion, or diffusion. In the simplest case, ventilation (breathing) in an atmosphere with low PO_2 (for example, at high altitude) will cause hypoxemia that can be corrected by breathing air with higher PO_2 . A second cause of hypoxemia is **hypoventilation**. With insufficient ventilation (for example, with shallow breathing), PA_{CO_2} rises and PA_{O_2} falls, due to inefficient exchange of alveolar air with the atmosphere. Consequently, hypoxemia and **hypercapnea** (high Pa_{CO_2}) result. Hypoxemia of this type can be corrected by adjusting respiratory rate and tidal volume, or breathing air containing a high concentration of oxygen.

Diffusion abnormalities are a third cause of hypoxemia. Poor diffusion (for example, due to thickening of the alveolar-capillary membrane) will produce hypoxemia with an elevated **alveolar-to-systemic arterial PO_2 gradient (A-a PO_2 gradient)**; this abnormal gradient occurs because PO_2 is not fully equilibrated between alveolar air and the blood leaving alveolar capillaries. Although hypoxemia will be corrected by administration of 100% oxygen, the A-a PO_2 gradient will remain elevated.

Shunt flow also produces hypoxemia. In the presence of shunt flow, venous admixture of oxygenated and deoxygenated blood produces an elevated A-a PO_2 gradient. In congenital abnormalities of the heart in which right-to-left shunting of blood occurs through atrial or ventricular septal defects, large A-a PO_2 gradients may be seen. Airway obstruction also results in shunt flow and increased A-a PO_2 gradient. Hypoxemia (low arterial PO_2) caused by shunt flow can be differentiated from hypoxemia caused by other defects by measurement of Pa_{O_2} before and after several minutes of breathing of 100% oxygen. Neither the hypoxemia nor the A-a PO_2 gradient associated with shunt flow can be fully corrected by breathing 100% oxygen, due to venous admixture of shunt flow to oxygenated blood.

Ventilation-perfusion imbalance is a fifth cause of low Pa_{O_2} . As discussed earlier, gradients in ventilation and perfusion produce some degree of ventilation-perfusion imbalance even in a healthy subject in the standing position, and regional ventilation-perfusion imbalances contribute to the normal, small A-a gradient PO_2 . If the imbalance is abnormally pronounced, mixing of blood from areas of the lung that are well perfused but underventilated with blood from other areas (where ventilation and perfusion are better balanced) causes an exaggerated A-a PO_2 gradient and hypoxemia. This type of hypoxemia can be corrected by administration of 100% oxygen.

A. Conditions with low ventilation/perfusion ratio



B. Conditions with high ventilation/perfusion ratio

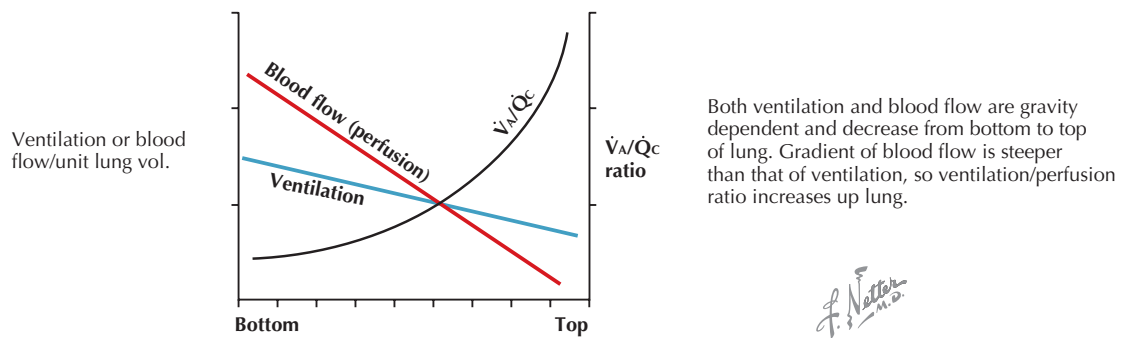
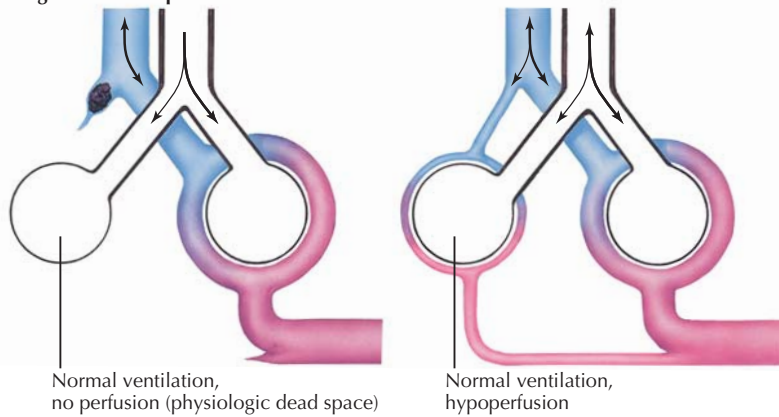


Figure 13.14 Ventilation-Perfusion ($\dot{V}_A/\dot{Q}_C$) Relationships In the standing position, the effects of gravity result in gradients in both perfusion and ventilation of the lung from base to apex. Because the perfusion gradient is steeper than the ventilation gradient, the ratio of ventilation to perfusion, $\dot{V}_A/\dot{Q}_C$, is lowest at the bottom of the lung and greatest at the top of the lung (B). $\dot{V}_A/\dot{Q}_C$ is also affected by various other conditions affecting ventilation and perfusion (A and B).

CHAPTER 14

The Mechanics of Breathing

The physical forces resulting in ventilation of the lungs are analogous to those creating blood flow in the cardiovascular system. A pressure gradient is required, and in the case of the lung, the gradient is created by movement of the chest wall and diaphragm. Flow of air occurs against the resistance of the airways, analogous to the resistance of blood vessels. However, factors affecting pressure, flow, and resistance are complex, and often different, in the two systems. This chapter details the physical forces involved in ventilation of the lungs and some of the changes that take place in disease.

BASIC MECHANICS OF THE VENTILATORY APPARATUS

Ventilation occurs as a result of mechanical forces associated with the chest wall and lungs. Both the lungs and the chest wall are **elastic**; that is, they passively recoil after being distended. **Elastic recoil pressure** is the pressure caused by distension. Functionally, the chest wall includes the diaphragm and abdominal muscles in addition to the rib cage. The **visceral pleura** (outer lining) of the lungs apposes the **parietal pleura** of the chest wall, and the small, fluid-filled space between the pleurae, the **pleural cavity**, contains only a few milliliters (mL) of fluid. The muscles of breathing are illustrated in Figure 14.1. The **diaphragm** is the major muscle for inspiration during normal, quiet breathing. As it contracts and its domes descend, the thoracic space is enlarged, decreasing alveolar pressure and resulting in inward flow of air through the airways. During more active ventilation, for example during exercise, the **intercostal muscles** have greater involvement in inspiration, elevating the ribs and expanding the chest as they contract. Expiration is a passive process during normal, quiet breathing and results from passive recoil of the lungs. During active breathing, various muscles of the abdominal wall, along with some of the intercostal muscles, contribute to the force, resulting in expiration.

Elastic Recoil of the Chest Wall and Lungs

The interactions between the forces of the chest wall and lungs during normal, quiet breathing are illustrated in Figure 14.2. In the context of elastic properties of the system at all lung volumes, from **functional residual capacity** (FRC; the volume of air in the lungs after normal, quiet expiration) to total lung



Mechanically, the lungs and chest wall can be conceptualized as a respiratory pump, acting in unison to produce pressures required to generate airflow. Flow is driven by the pressure gradient between the alveolar space and the mouth opening, produced by the elastic properties of the system and activity of respiratory muscles. An additional characteristic of this pump is airway resistance (R_{aw}), defined by the formula:

$$R_{aw} = \frac{(P_A - P_{ATM})}{\text{rate of airflow}}$$

where P_A is alveolar pressure and P_{ATM} is atmospheric pressure. This formula can be rearranged to:

$$\text{rate of airflow} = \frac{(P_A - P_{ATM})}{R_{aw}}$$

This formula is analogous to the formula for blood flow ($Q = \Delta P/R$, where ΔP is the pressure gradient along which flow occurs, and R is the resistance to flow).

capacity (TLC), chest wall and lung forces are illustrated in Figure 14.3. At FRC, the mechanical system is at rest, and the pressure within the airways and alveolar space is equal to atmospheric pressure (see Fig. 14.2A). In this state, muscles of the chest wall are relaxed, and the outward elastic recoil pressure of the chest wall is equal and opposite to the inward elastic recoil pressure of the lungs. In other words, at FRC, although the tendency of the chest wall is to expand and the tendency of the lungs is to collapse, these two forces are in balance, keeping the volume of the lungs at FRC. The algebraic sum of the negative recoil pressure created by the chest wall and the positive recoil pressure of the lungs is zero; pleural pressure (pressure in the pleural cavity) is negative (subatmospheric). Note the balance between chest wall recoil and lung recoil pressures at FRC in Figure 14.3.

Forces during Inspiration and Expiration

Inspiration occurs when the rib cage expands and the diaphragm moves downward, creating negative pressure within the lungs (see Fig. 14.2B). Pleural pressure is more negative than in the resting state, as the chest wall exerts greater outward pressure. Air flows into the lung until alveolar pressure reaches atmospheric pressure. At the end of inspiration, during normal, quiet breathing (tidal volume of 500 mL), chest wall